SCOOT group quizzes

Six quizzes of ten multiple-choice questions for each of the six investigation groups in the SCOOT Foundation lesson (360 questions). Quiz 1 covers the basics, quiz 2 key words, quiz 3 numbers and calculations, quiz 4 “what happens if…?”, quiz 5 everyday pictures and quiz 6 is an exam-style challenge. Each quiz takes about 10 minutes. Answers with short explanations are at the end.

Group	Topic	Builds on
Group 1	Small nudges (the elastic plan)	F1 · F6 · F7 · F8
Group 2	Detectors and LPUs	F2 · F3
Group 3	Platoons spread	F2 · F3 · F7
Group 4	Queue and degree of saturation	F1 · F3 · F4 · F5 · F6
Group 5	Validation	F2 · F4 · F6
Group 6	SCOOT vs others	F1 · F5 · F8

Interactive versions with instant feedback: smrahmanbd.github.io/scoot-learning-lab/quiz.html

GROUP 1 · SMALL NUDGES (THE ELASTIC PLAN) · QUIZ 1 OF 6

Basics

Names: _____ Score: ____ / 10

1. **What are the three timing "dials" in SCOOT's name?**

   A ☐ Speed, flow and density
   B ☐ Phase, stage and link
   C ☐ Red, amber and green
   D ☐ Split, cycle and offset

2. **What is the split?**

   A ☐ How the green time is shared between the stages at a junction
   B ☐ The time to run every stage once
   C ☐ The amber plus all-red time
   D ☐ The time between greens at neighbouring junctions

3. **What is the offset?**

   A ☐ The difference between two splits
   B ☐ The time a car takes to cross the junction
   C ☐ The length of the amber
   D ☐ The time from green at one junction to green at the next one

4. **How often does SCOOT's split optimiser make a decision?**

   A ☐ Once an hour
   B ☐ Every 5 minutes
   C ☐ Once a day
   D ☐ Before every stage change

5. **How often is the cycle time of a region normally reviewed?**

   A ☐ Every second
   B ☐ At every stage change
   C ☐ Every 5 minutes (2.5 when traffic is rising)
   D ☐ Once a week

6. **A junction has two stages and a 60 s cycle. How many split decisions does SCOOT make in one hour?**

   A ☐ 120
   B ☐ 60
   C ☐ 12
   D ☐ 192

7. **Add 120 split decisions, 60 offset decisions (one per cycle) and 12 cycle reviews (every 5 minutes). How many decisions per hour?**

   A ☐ 192
   B ☐ 240
   C ☐ 180
   D ☐ 132

8. **Why does SCOOT change the timings by only a few seconds at a time?**

   A ☐ So drivers do not notice the change
   B ☐ The controller cannot make bigger changes
   C ☐ One wrong guess cannot do much harm, and many small nudges follow the traffic smoothly
   D ☐ To save electricity

9. **Which everyday object is the best picture of the elastic plan?**

   A ☐ A thermostat that nudges the heating every few minutes
   B ☐ A light switch
   C ☐ A printed bus timetable
   D ☐ A road map

10. **How is SCOOT's elastic plan different from "plan selection"?**

    A ☐ SCOOT lets every junction pick its own cycle freely
    B ☐ SCOOT stretches the amber when traffic is heavy
    C ☐ SCOOT edits the running plan a little at a time instead of swapping whole stored plans
    D ☐ SCOOT swaps between stored plans at fixed times of day

GROUP 1 · SMALL NUDGES (THE ELASTIC PLAN) · QUIZ 2 OF 6

Key words

Names: _____ Score: ____ / 10

1. **What is the cycle C of a signal?**

   A ☐ The time to run every stage once

   B ☐ The time between greens at two neighbouring junctions

   C ☐ The share of green given to the main road

   D ☐ The amber time at the end of a green

2. **What is a stage?**

   A ☐ A group of movements that get green together

   B ☐ A signalled junction

   C ☐ One approach road from detector to stop line

   D ☐ The time between one car and the next

3. **In SCOOT, what is a region?**

   A ☐ One lane of one approach road

   B ☐ The area covered by one loop detector

   C ☐ A group of junctions that all share one cycle time

   D ☐ The stretch of road between the loop and the stop line

4. **What is the intergreen?**

   A ☐ The green time given to the side road

   B ☐ The amber plus all-red between two stages

   C ☐ The time a car takes to cross the loop

   D ☐ The gap between two cars in a platoon

5. **In SCOOT words, what is a link?**

   A ☐ One approach road, from its detector to its stop line

   B ☐ The cable joining two signal poles

   C ☐ The time from green at one junction to green at the next

   D ☐ A group of junctions sharing a cycle

6. **SCOOT is called centralised. What does that mean?**

   A ☐ The loops sit in the centre of the junction

   B ☐ Each junction decides its own timings on its own

   C ☐ One computer in the control centre runs the model and sets the timings for every junction

   D ☐ Only the junction in the town centre is controlled

7. **SCOOT is called model-based. What does that mean?**

   A ☐ It only works in a computer simulation

   B ☐ It copies the timings of a model junction

   C ☐ The model's predictions, not the raw detector counts, decide the timings

   D ☐ It picks a model plan from a library each morning

8. **What does the phrase elastic plan mean?**

   A ☐ The plan is thrown away and rebuilt every second

   B ☐ Drivers can stretch the green by pressing a button

   C ☐ The amber stretches when the road is wet

   D ☐ The timings stretch a little at a time to fit demand, instead of jumping between stored plans

9. **What is a green wave?**

   A ☐ A long green given to buses only

   B ☐ Neighbouring signals timed so a platoon released by one meets a green at the next

   C ☐ All signals in town turning green at the same moment

   D ☐ A green light that flashes before turning amber

10. **Which SCOOT optimiser works on the slowest clock?**

- **A** ☐ The offset optimiser (every cycle)
- **B** ☐ The cycle optimiser (every 5 minutes)
- **C** ☐ They all work at the same speed
- **D** ☐ The split optimiser (every stage change)

GROUP 1 · SMALL NUDGES (THE ELASTIC PLAN) · QUIZ 3 OF 6

Numbers and calculations

Names: _____ Score: ____ / 10

1. **A junction runs a 60 s cycle. How many offset decisions does SCOOT make in one hour?**
   - **A** ☐ 12
   - **B** ☐ 120
   - **C** ☐ 60
   - **D** ☐ 192

2. **A junction has three stages and an 80 s cycle. How many split decisions does SCOOT make in an hour?**
   - **A** ☐ 240
   - **B** ☐ 45
   - **C** ☐ 90
   - **D** ☐ 135

3. **While traffic is rising, the region's cycle is reviewed every 2.5 minutes. How many cycle decisions is that per hour?**
   - **A** ☐ 25
   - **B** ☐ 24
   - **C** ☐ 12
   - **D** ☐ 150

4. **A two-stage junction runs a 90 s cycle, and the cycle is reviewed every 5 minutes. How many timing decisions in total per hour (split + offset + cycle)?**
   - **A** ☐ 120
   - **B** ☐ 92
   - **C** ☐ 192
   - **D** ☐ 132

5. **A 60 s cycle has green A = 26 s and two intergreens of 5 s each. How long is green B?**
   - **A** ☐ 34 s
   - **B** ☐ 29 s
   - **C** ☐ 24 s
   - **D** ☐ 19 s

6. **A junction has four stages and a 100 s cycle. How many split decisions does SCOOT make in an hour?**
   - **A** ☐ 144
   - **B** ☐ 400
   - **C** ☐ 36
   - **D** ☐ 72

7. **Two junctions are 300 m apart and the traffic travels at 15 m/s. About what offset gives a green wave?**
   - **A** ☐ 0.05 s
   - **B** ☐ 20 s
   - **C** ☐ 4500 s
   - **D** ☐ 45 s

8. **Two junctions are 400 m apart and traffic travels at 10 m/s. About what offset should SCOOT walk towards?**
   - **A** ☐ 4000 s
   - **B** ☐ 25 s
   - **C** ☐ 4 s
   - **D** ☐ 40 s

9. **The stored green is 22 s. At three stage changes in a row the split optimiser picks the −4 s option, and each time the stored plan moves by 1 s. What is the stored green now?**
   - **A** ☐ 10 s
   - **B** ☐ 18 s
   - **C** ☐ 21 s
   - **D** ☐ 19 s

10. **A 48 s cycle loses 12 s per cycle to stage changes. What share of the cycle is lost time?**
    - **A** ☐ 4%
    - **B** ☐ 25%
    - **C** ☐ 36%
    - **D** ☐ 12%

GROUP 1 · SMALL NUDGES (THE ELASTIC PLAN) · QUIZ 4 OF 6

What happens if…?

Names: _____ Score: ____ / 10

1. **During the morning rush, more and more cars arrive on the main road. What does SCOOT do to its split?**
   - **A** ☐ It gives the main-road stage a few more seconds of green, step by step
   - **B** ☐ It leaves the split alone and only changes the offset
   - **C** ☐ It gives all the extra green to the side road
   - **D** ☐ It switches to a stored "rush hour" plan at 8:00

2. **The cycle time is lengthened from 60 s to 90 s. What happens to the number of split and offset decisions per hour?**
   - **A** ☐ They fall, because there are fewer cycles per hour
   - **B** ☐ They drop to zero until the next cycle review
   - **C** ☐ They rise, because each cycle is longer
   - **D** ☐ They stay the same, because the clock is fixed

3. **Traffic in the region starts rising quickly. What happens to how often the cycle is reviewed?**
   - **A** ☐ It is reviewed more often: every 2.5 minutes instead of 5
   - **B** ☐ It is reviewed less often, to keep the plan stable
   - **C** ☐ It is frozen until traffic settles
   - **D** ☐ It is reviewed at every stage change

4. **The busiest (critical) junction in a region rises above about 90% degree of saturation. What does the cycle optimiser do?**
   - **A** ☐ It shortens the cycle of that junction only
   - **B** ☐ It switches the junction to flashing amber
   - **C** ☐ It lengthens the cycle of the whole region
   - **D** ☐ It removes one stage

5. **SCOOT lengthens the cycle while the lost time per cycle stays 12 s. What happens to the share of time lost to stage changes?**
   - **A** ☐ It gets smaller
   - **B** ☐ It gets bigger
   - **C** ☐ It stays exactly the same
   - **D** ☐ It becomes zero

6. **A fixed plan keeps a 24 s green all morning, but the rush brings more traffic than 24 s can clear. What happens?**
   - **A** ☐ The green stretches by itself
   - **B** ☐ Nothing changes, because the plan was designed for the average
   - **C** ☐ The queue clears faster because drivers hurry
   - **D** ☐ A queue piles up and grows every cycle

7. **The peak eases and the critical junction drops well below 90%. What does the cycle optimiser do?**
   - **A** ☐ It holds or shortens the cycle
   - **B** ☐ It stops the split optimiser
   - **C** ☐ It doubles the offset
   - **D** ☐ It lengthens the cycle even more

8. **The offset between two junctions moves closer to the travel time between them. What happens to stops at the second junction?**
   - **A** ☐ They stay the same, because offsets only affect the first junction
   - **B** ☐ The second junction's cycle doubles
   - **C** ☐ They fall, because the platoon now meets a green
   - **D** ☐ They rise, because cars arrive too early

9. **SCOOT's guess about the traffic is wrong for one decision. Why does little harm follow?**
   - **A** ☐ The change was only a few seconds, and the next decision soon corrects it
   - **B** ☐ SCOOT never makes wrong guesses
   - **C** ☐ The drivers ignore the change
   - **D** ☐ The controller undoes all changes at midnight

10. **At one stage change the split optimiser finds the +4 s option best. What happens to the stored plan?**

A ☐ It moves by the full 4 s

B ☐ It moves by only 1 s in that direction

C ☐ It is replaced by a new stored plan

D ☐ It does not change at all

GROUP 1 · SMALL NUDGES (THE ELASTIC PLAN) · QUIZ 5 OF 6

Everyday pictures

Names: _____ Score: ____ / 10

1. **The course compares SCOOT to a thermostat. Where does that picture break down?**
   - **A** ☐ A thermostat has one knob; SCOOT turns three knobs, each at its own speed, at every junction
   - **B** ☐ A thermostat changes once a year
   - **C** ☐ A thermostat uses a model; SCOOT uses only raw counts
   - **D** ☐ A thermostat makes small changes; SCOOT makes big ones

2. **The thermostat "nudges the heating a little every few minutes". What is the opposite picture in the analogy?**
   - **A** ☐ A heater with three dials
   - **B** ☐ A timer that switches between "winter" and "summer" settings
   - **C** ☐ A window left open
   - **D** ☐ A thermostat that nudges every second

3. **At the memory-palace town gate, a giant rubber band is stretched around the traffic light. What does the rubber band stand for?**
   - **A** ☐ The loop detector
   - **B** ☐ The region boundary
   - **C** ☐ The elastic plan
   - **D** ☐ The queue at the stop line

4. **In the same scene, the tiny engineer twangs the band every few seconds. What does that show?**
   - **A** ☐ SCOOT changes the plan once a day
   - **B** ☐ The signals break down often
   - **C** ☐ SCOOT makes small, frequent changes
   - **D** ☐ Drivers keep pressing buttons

5. **A signal is described as a teacher letting one row of students out at a time. In that picture, what is one row?**
   - **A** ☐ One car
   - **B** ☐ One stage: the movements that get green together
   - **C** ☐ One loop detector
   - **D** ☐ One whole cycle

6. **The course pictures three runners: a robot with a printed timetable, a jumpy rabbit and a calm dispatcher with headphones. Which one is SCOOT?**
   - **A** ☐ The rabbit, finishing at 14 and a half
   - **B** ☐ The calm dispatcher, finishing just under 13
   - **C** ☐ None of them; SCOOT is the finish line
   - **D** ☐ The robot, finishing at 19

7. **In the elastic-plan animation (morning rush), what happens under the fixed 24 s green?**
   - **A** ☐ The green grows with the traffic
   - **B** ☐ The queue shrinks as the rush grows
   - **C** ☐ A queue piles up because the rush brings more traffic than 24 s can clear
   - **D** ☐ The light stays red all morning

8. **A cycle bar shows: Green A 26 s, Int. 5, Green B 24 s, Int. 5. What does the whole length of the bar stand for?**
   - **A** ☐ The cycle, 60 s
   - **B** ☐ The split, 26 s
   - **C** ☐ The lost time, 10 s
   - **D** ☐ The offset to the next junction

9. **In the SCOOT loop diagram, what sits between the detector and the three optimisers?**
   - **A** ☐ The link model: profile, queue and DoS
   - **B** ☐ The drivers
   - **C** ☐ The stop-line push button
   - **D** ☐ A library of stored plans

10. **The offset animation is a time–distance picture of two junctions 300 m apart. What does each sloping line show?**
    - **A** ☐ One cycle of the signal
    - **B** ☐ One loop detector
    - **C** ☐ One traffic light
    - **D** ☐ One car moving along the road over time

GROUP 1 · SMALL NUDGES (THE ELASTIC PLAN) · QUIZ 6 OF 6

Exam challenge

Names: _____ Score: ____ / 10

1. **A two-stage junction runs a 60 s cycle during a rising peak, when the cycle is reviewed every 2.5 minutes. How many timing decisions per hour?**

   A ☐ 204
   B ☐ 192
   C ☐ 216
   D ☐ 180

2. **Which line matches each knob to its clock correctly?**

   A ☐ All three: once a day
   B ☐ Split: every cycle · Offset: every 5 min · Cycle: every stage change
   C ☐ Split: every stage change · Offset: every cycle · Cycle: every 5 min
   D ☐ Split: every 5 min · Offset: every stage change · Cycle: every cycle

3. **Why must neighbouring junctions in a region share one cycle time?**

   A ☐ Otherwise their greens drift apart and the offset (green wave) cannot be kept
   B ☐ So that every junction has the same split
   C ☐ So the detectors can be read 4 times a second
   D ☐ Because the central computer can store only one number

4. **An approach has s = 1800 veh/h, green g = 25 s and cycle C = 60 s, with a demand of 540 veh/h. What is its DoS, and is the green big enough?**

   A ☐ 0.30; far too much green
   B ☐ 0.72; no, it is above 1
   C ☐ 0.72; yes, it is below the ~90% target
   D ☐ 1.39; no, the queue grows

5. **During a peak, flow ratio Y = 0.78 and lost time L = 12 s. Using DoS = Y × C ÷ (C − L), what is the DoS when SCOOT lengthens the cycle to 64 s?**

   A ☐ 0.96
   B ☐ 0.78
   C ☐ 0.81
   D ☐ 1.07

6. **Before a stage change the three trial options give worst DoS: −4 s 0.92, no change 0.80, +4 s 0.71. The stored green is 30 s. What happens?**

   A ☐ −4 s is used, because 0.92 is closest to 90%
   B ☐ No change, because 0.80 is already below 90%
   C ☐ +4 s is used and the stored green becomes 34 s
   D ☐ +4 s is used now and the stored green becomes 31 s

7. **In Lab F3, mean delay was fixed 18.96 s and SCOOT 12.79 s. About how much less delay did SCOOT give than fixed time?**

   A ☐ About 6 s more
   B ☐ About 48%
   C ☐ About 12%
   D ☐ About 33%

8. **How does SCOOT differ from actuated control? (links to group 6)**

   A ☐ There is no difference; both follow a stored plan
   B ☐ SCOOT uses upstream loops and a model to adapt split, cycle and offset across a region; actuated extends the green from stop-line loops, locally
   C ☐ Actuated changes the cycle of the whole region; SCOOT only one junction
   D ☐ SCOOT uses stop-line loops; actuated uses upstream loops

9. **Why is the SCOOT detector placed upstream, not at the stop line? (links to group 2)**

   A ☐ Because stop-line loops cannot detect metal
   B ☐ So the loop counts only parked cars
   C ☐ So the loop is read only once per cycle
   D ☐ It gives 10–20 s of warning, so the model can prepare the timings before the platoon arrives

10. **On a corridor run, SCOOT's mean delay was 30.4 s against actuated 29.1 s, but stops fell from 1.15 to 1.10 and bus delay from 38.0 to 22.9 s. What is the fair conclusion?**

A ☐ SCOOT was better in every way

B ☐ SCOOT is not better on every measure, but it gave fewer stops and much less bus delay

C ☐ The numbers prove the detectors were broken

D ☐ SCOOT was worse in every way

GROUP 2 · DETECTORS AND LPUS · QUIZ 1 OF 6

Basics

Names: _____ Score: ____ / 10

1. **Where is the SCOOT detector loop normally placed?**
   - **A** ☐ Near the far end of the link, just after the upstream junction
   - **B** ☐ In the middle of the junction
   - **C** ☐ On the exit road
   - **D** ☐ At the stop line

2. **What does SCOOT gain from putting the loop there?**
   - **A** ☐ About 10–20 s of warning before the cars reach the stop line
   - **B** ☐ An exact speed for every car
   - **C** ☐ A cheaper installation
   - **D** ☐ A count of pedestrians

3. **How often is a SCOOT loop read?**
   - **A** ☐ Once a cycle
   - **B** ☐ 4 times a second
   - **C** ☐ Every 5 minutes
   - **D** ☐ Once a second

4. **What does LPU stand for?**
   - **A** ☐ Loop pulse unit
   - **B** ☐ Lost period unit
   - **C** ☐ Link profile unit
   - **D** ☐ Lane priority unit

5. **Which rule does this lab use to turn a vehicle into LPUs?**
   - **A** ☐ LPU = 1 × vehicle + 2 × seconds the loop is covered
   - **B** ☐ LPU = vehicles ÷ seconds
   - **C** ☐ LPU = 2 × vehicle + 1 × seconds covered
   - **D** ☐ LPU = seconds covered only

6. **A 4.5 m car crosses a 2 m loop at 13.5 m/s. How long is the loop covered?**
   - **A** ☐ 0.33 s
   - **B** ☐ 1.04 s
   - **C** ☐ 0.48 s
   - **D** ☐ 2.25 s

7. **Using the lab's rule, how many LPUs is that car?**
   - **A** ☐ 1.00
   - **B** ☐ 2.96
   - **C** ☐ 1.48
   - **D** ☐ 1.96

8. **A 12 m bus crosses the same loop at the same speed. How many LPUs?**
   - **A** ☐ 1.04
   - **B** ☐ 2.07
   - **C** ☐ 4.00
   - **D** ☐ 3.07

9. **In a jam, a car crawls slowly over the loop. What happens to its LPU value?**
   - **A** ☐ It falls
   - **B** ☐ It becomes zero
   - **C** ☐ It stays exactly the same
   - **D** ☐ It rises, because the loop is covered for longer

10. **Why does SCOOT use LPUs instead of plain vehicle counts?**
    - **A** ☐ The law requires them
    - **B** ☐ LPUs are easier to count
    - **C** ☐ LPUs ignore buses
    - **D** ☐ LPUs reflect how much green the traffic needs: long or slow vehicles count more

GROUP 2 · DETECTORS AND LPUS · QUIZ 2 OF 6

Key words

Names: _____ Score: ____ / 10

1. **What is a loop detector?**

   A ☐ A coil of wire under the road surface that switches “on” when metal passes over it

   B ☐ A radio that talks to each driver’s phone

   C ☐ A pressure plate that weighs each vehicle

   D ☐ A camera on the signal pole that films the queue

2. **In SCOOT, what does occupancy describe?**

   A ☐ How long the loop is covered by vehicles

   B ☐ How long the green light lasts

   C ☐ How many lanes the road has

   D ☐ How many people are in each car

3. **In SCOOT words, what is a link?**

   A ☐ The time between two greens

   B ☐ One approach road, from its detector to its stop line

   C ☐ The cable from the loop to the controller

   D ☐ A whole town’s network of signals

4. **In SCOOT words, what is a node?**

   A ☐ A single car

   B ☐ A signalled junction

   C ☐ A stage of green

   D ☐ A detector loop

5. **What is a platoon?**

   A ☐ A line of parked cars

   B ☐ A row of loops across all lanes

   C ☐ A bunch of cars released together by the upstream green

   D ☐ A group of signals sharing one cycle

6. **What is the cruise time T of a link?**

   A ☐ How long the loop is covered by one car

   B ☐ How long the green lasts

   C ☐ How long a car takes from the detector to the stop line

   D ☐ How long one whole cycle lasts

7. **What does the word upstream mean for a SCOOT detector?**

   A ☐ Uphill from the junction

   B ☐ On the side road only

   C ☐ Towards where traffic comes from: just after the previous junction

   D ☐ Beyond the stop line, on the exit road

8. **What is flow q?**

   A ☐ The share of time the loop is covered

   B ☐ The time between two cars

   C ☐ The length of the queue

   D ☐ The number of vehicles passing per hour (or per second)

9. **What is a headway?**

   A ☐ The distance from loop to stop line

   B ☐ The seconds between one car and the next

   C ☐ The front half of a platoon

   D ☐ The length of a car

10. **Which kind of signal control usually puts its loops at the stop line?**

    A ☐ SCOOT

    B ☐ Webster’s method

    C ☐ Actuated control

    D ☐ Fixed time

GROUP 2 · DETECTORS AND LPUS · QUIZ 3 OF 6

Numbers and calculations

Names: _____ Score: ____ / 10

1. **A 4 m car crosses a 2 m loop at 12 m/s. Using the lab rule, how many LPUs?**
   - **A** ☐ 1.00
   - **B** ☐ 1.50
   - **C** ☐ 2.00
   - **D** ☐ 1.67

2. **An 8 m van crosses a 2 m loop at 5 m/s. How long is the loop covered?**
   - **A** ☐ 1.6 s
   - **B** ☐ 0.5 s
   - **C** ☐ 50 s
   - **D** ☐ 2.0 s

3. **The course's 16.5 m lorry crosses the 2 m loop at 9.25 m/s. How many LPUs?**
   - **A** ☐ 4.57
   - **B** ☐ 2.0
   - **C** ☐ 5.0
   - **D** ☐ 3.0

4. **At 13.5 m/s the course's car is 1.96 LPU and the bus 3.07 LPU. How many "cars" is the bus worth?**
   - **A** ☐ About 1.11
   - **B** ☐ About 0.64
   - **C** ☐ About 2.67
   - **D** ☐ About 1.57

5. **The validation lab measured about 1.95 LPU per vehicle. About how many LPUs is a queue of 10 such vehicles?**
   - **A** ☐ 39 LPU
   - **B** ☐ 19.5 LPU
   - **C** ☐ 10 LPU
   - **D** ☐ 9.75 LPU

6. **A loop is 180 m upstream of the stop line and cars drive at 12 m/s. How many seconds of warning does SCOOT get?**
   - **A** ☐ 0.07 s
   - **B** ☐ 168 s
   - **C** ☐ 15 s
   - **D** ☐ 2160 s

7. **The loop is read 4 times a second. A vehicle covers it for 1.5 s. How many readings say "on"?**
   - **A** ☐ 1.5
   - **B** ☐ 0.375
   - **C** ☐ 6
   - **D** ☐ 4

8. **In one second, two cars pass the loop and together cover it for 0.5 s. How many LPUs for that second?**
   - **A** ☐ 3.0
   - **B** ☐ 6.0
   - **C** ☐ 2.0
   - **D** ☐ 2.5

9. **A 4.5 m car crosses the 2 m loop at 36 km/h. How many LPUs?**
   - **A** ☐ 1.65
   - **B** ☐ 2.30
   - **C** ☐ 1.90
   - **D** ☐ 1.36

10. **The course's 12 m bus crawls over the 2 m loop at 6.75 m/s (half of 13.5 m/s). How many LPUs?**
    - **A** ☐ 6.14
    - **B** ☐ 2.07
    - **C** ☐ 5.15
    - **D** ☐ 3.07

GROUP 2 · DETECTORS AND LPUS · QUIZ 4 OF 6

What happens if…?

Names: _____ Score: ____ / 10

1. **A car crosses the loop twice as fast as before. What happens to its covered time and LPU?**
   - **A** ☐ Covered time doubles, so its LPU rises
   - **B** ☐ Covered time halves, so its LPU falls towards 1
   - **C** ☐ Both stay the same
   - **D** ☐ Its LPU doubles

2. **A town fits 4 m loops instead of 2 m loops, with the same traffic. What happens to each vehicle's LPU?**
   - **A** ☐ It becomes exactly 2 for every vehicle
   - **B** ☐ It rises, because each vehicle covers the longer loop for longer
   - **C** ☐ It falls, because the loop is less sensitive
   - **D** ☐ It stays the same, because LPU depends only on the vehicle

3. **The SCOOT loop is moved from upstream to the stop line. What is the main loss?**
   - **A** ☐ The loop is read less often
   - **B** ☐ The warning: arrivals are seen too late and the loop is covered by the standing queue
   - **C** ☐ Buses stop scoring more than cars
   - **D** ☐ The loop can no longer count cars at all

4. **The detector counted only vehicles and ignored how long it was covered. What would go wrong?**
   - **A** ☐ SCOOT would give buses too much green
   - **B** ☐ A bus and a car would look the same, so the green needed by long vehicles would be underestimated
   - **C** ☐ The loop would be read more often
   - **D** ☐ Nothing; counts and LPUs are the same thing

5. **At the same speed, a 4.5 m car is replaced by a 7 m van. What happens to the LPU?**
   - **A** ☐ It rises from about 1.96 to about 2.33
   - **B** ☐ It falls from about 1.96 to about 1.50
   - **C** ☐ It stays at 1.96
   - **D** ☐ It jumps to about 3.74

6. **A loop ends up only 40 m from the stop line, giving about 3 s of cruise time. What happens to SCOOT's planning?**
   - **A** ☐ The LPU rule changes to 2 × vehicles
   - **B** ☐ Nothing changes; distance does not matter
   - **C** ☐ It gets far less warning than the usual 10–20 s, so it has less time to act
   - **D** ☐ It gets more warning, because cars arrive sooner

7. **Through the morning the same number of vehicles arrive, but more of them are long lorries. What happens to the LPUs per cycle?**
   - **A** ☐ They fall, because lorries are slower to count
   - **B** ☐ They rise, so the model sees more demand and needs more green
   - **C** ☐ They drop to zero for lorries
   - **D** ☐ They stay the same, because the count is the same

8. **The speed limit on a link drops, and the loop stays in the same place. What happens?**
   - **A** ☐ Cruise time T grows (more warning) and each car covers the loop longer (more LPUs)
   - **B** ☐ Nothing changes
   - **C** ☐ T shrinks and LPUs fall
   - **D** ☐ Only the count changes

9. **A student compares a queue of 10 cars with a model queue of 20 LPU and says "the model doubled the queue". What has gone wrong?**
   - **A** ☐ Nothing is doubled: 10 cars are about 20 LPU, so the two agree
   - **B** ☐ The model has a fault
   - **C** ☐ The loop counted every car twice
   - **D** ☐ SCOOT always overestimates queues

10. **A bus slows from 13.5 m/s to 9 m/s over the loop. Which part of its LPU changes?**

A ☐ Only the “1 × vehicle” part, which grows

B ☐ Neither part changes

C ☐ Only the “2 × seconds covered” part, which grows

D ☐ Both parts halve

GROUP 2 · DETECTORS AND LPUS · QUIZ 5 OF 6

Everyday pictures

Names: _____ Score: ____ / 10

1. **In the doorman picture, every person pays one coin just to enter. What does that coin stand for in the LPU rule?**

   A ☐ The green time
   B ☐ The seconds the loop is covered
   C ☐ The "1 × vehicle" count
   D ☐ The 2 m loop length

2. **The doorman also charges two coins for every second someone blocks the door. What is that in SCOOT terms?**

   A ☐ Reading the loop 4 times a second
   B ☐ The 10–20 s warning
   C ☐ 2 × number of vehicles
   D ☐ 2 × seconds the loop is occupied

3. **Why does someone pushing a wide trolley pay about three coins, while a quick walker pays a little over one?**

   A ☐ The trolley blocks the doorway longer, just as a bus covers the loop longer
   B ☐ Trolleys are counted as three people
   C ☐ The walker gets a discount for speed
   D ☐ The doorman dislikes trolleys

4. **Where does the doorman picture "break", in a way that turns out useful?**

   A ☐ The loop cannot tell metal from people
   B ☐ The doorman only works at night
   C ☐ The doorman never charges buses
   D ☐ Slow traffic is also charged more, because it stays on the loop longer, and slow traffic usually means queueing

5. **In the memory palace, the toll doormat lies far from the lights, just past the gate. What does that position remind you of?**

   A ☐ The doormat marks the end of the cycle
   B ☐ The doormat is the traffic light
   C ☐ The detector sits at the upstream end of the link
   D ☐ The detector sits at the stop line

6. **In the bus-stop picture, SCOOT is the dispatcher with a radio who hears "a crowd is coming in 15 seconds". What gives SCOOT that radio message?**

   A ☐ The driver pressing a button
   B ☐ The upstream loop, which counts the platoon before it reaches the stop line
   C ☐ A camera watching the queue
   D ☐ A printed timetable

7. **The upstream-detection animation shows a platoon from the upstream green crossing the loop and then reaching the next junction. What is the main point?**

   A ☐ Cars are counted only when they stop
   B ☐ The loop changes the lights the moment a car crosses it
   C ☐ The platoon is counted long before the stop line, so it can arrive on green
   D ☐ The platoon gets shorter as it travels

8. **The slow-motion LPU animation shows a 4.5 m car and a 12 m bus crossing a 2 m loop at 13.5 m/s. What should you notice?**

   A ☐ The car scores more because it is faster
   B ☐ The bus keeps the loop covered about twice as long, so it scores about 3.07 LPU against 1.96
   C ☐ Both score exactly 1 LPU
   D ☐ The loop misses the bus

9. **The bar chart shows LPU per vehicle at 13.5 m/s: car 1.96, van 2.33, bus 3.07, artic lorry 3.74. What pattern does it show?**

A ☐ LPUs fall as vehicles get longer

B ☐ Only buses score more than 2

C ☐ LPUs grow with vehicle length, weighting each vehicle by the green it uses

D ☐ All vehicles score the same

10. **A timeline of loop "on" blocks shows four blocks of about 0.5 s and one of about 1.05 s. What most likely made the wide block?**

A ☐ A long vehicle, such as a bus, covering the loop longer

B ☐ A gap with no traffic

C ☐ Two cars side by side in different lanes

D ☐ A broken loop

GROUP 2 · DETECTORS AND LPUS · QUIZ 6 OF 6

Exam challenge

Names: _____ Score: ____ / 10

1. **The lab's car scores 1.96 LPU and the SUMO validation lab measured 1.95 LPU per vehicle on average. What does this suggest?**

   A ☐ The LPU rule was ignored in SUMO

   B ☐ The traffic in the lab was mostly car-like vehicles near that speed

   C ☐ Every vehicle in the lab was a bus

   D ☐ Occupancy was zero in the lab

2. **Which statement correctly contrasts actuated control and SCOOT?**

   A ☐ Actuated loops at the stop line extend green while cars come; SCOOT's upstream loops feed a model that predicts

   B ☐ Both put loops upstream and use LPUs

   C ☐ SCOOT loops are at the stop line; actuated loops are upstream

   D ☐ Actuated control uses a model; SCOOT only reacts to gaps

3. **Which two things make a vehicle score more LPUs on a given loop?**

   A ☐ Being heavier and louder

   B ☐ Being shorter and going faster

   C ☐ Having more passengers and lights on

   D ☐ Being longer and going slower

4. **In one cycle, 12 cars and 2 buses cross the loop at 13.5 m/s. About how many LPUs is that?**

   A ☐ 23.52

   B ☐ 28

   C ☐ 29.66

   D ☐ 14

5. **Linking to group 3: a link has cruise time T = 10 s. With β = 0.8, when does the front of a platoon reach the stop line after crossing the loop?**

   A ☐ About 8 s later (lag βT)

   B ☐ Exactly 10 s later

   C ☐ About 2 s later

   D ☐ About 12.5 s later

6. **Linking to group 4: 9 cars (2 LPU each) arrive per cycle, no residual queue, effective green 25 s, SATO 0.8 LPU/s. What is the DoS?**

   A ☐ 1.11

   B ☐ 0.90

   C ☐ 0.45

   D ☐ 0.72

7. **A student writes: "The queue is 10 cars and SATO is 0.67 LPU/s, so it clears in 10 ÷ 0.67 ≈ 15 s." What is the main mistake?**

   A ☐ SATO should be in veh/h

   B ☐ The loop should be read once a second

   C ☐ Mixing units: the queue must be in LPUs (about 19.5), not vehicles

   D ☐ Queues never clear

8. **A 4.5 m car on the 2 m loop scores exactly 3.0 LPU. How fast was it going?**

   A ☐ 4.5 m/s

   B ☐ 13.5 m/s

   C ☐ 6.5 m/s

   D ☐ 3.25 m/s

9. **A lane carries 540 veh/h, all cars of about 1.96 LPU. Roughly how many LPU per second pass the loop?**

   A ☐ About 1058 LPU/s

   B ☐ About 0.29 LPU/s

   C ☐ About 0.15 LPU/s

   D ☐ About 6.7 LPU/s

10. **In Lab F3, SCOOT had lower mean delay than actuated control (12.79 s against 14.48 s). Which detector idea helps explain this?**

A ☐ SCOOT counts only buses

B ☐ Actuated loops cannot detect cars

C ☐ SCOOT's upstream loops give warning, so it can prepare the green before the platoon arrives

D ☐ SCOOT reads its loops less often

GROUP 3 · PLATOONS SPREAD · QUIZ 1 OF 6

Basics

Names: _____ Score: ____ / 10

1. **What is a platoon?**
   - **A** ☐ A convoy of buses only
   - **B** ☐ A single lane of a road
   - **C** ☐ The queue at the stop line
   - **D** ☐ A bunch of cars released together by the upstream green

2. **What is a cyclic flow profile?**
   - **A** ☐ The speed of every car
   - **B** ☐ The LPUs seen in each second of the cycle, averaged over several cycles
   - **C** ☐ The order of the stages
   - **D** ☐ A chart of traffic over a whole day

3. **Why does a platoon spread out on its way to the stop line?**
   - **A** ☐ Drivers travel at different speeds
   - **B** ☐ The signals change colour
   - **C** ☐ The road gets narrower
   - **D** ☐ The detector misses some cars

4. **What is the cruise time T?**
   - **A** ☐ The time to clear the queue
   - **B** ☐ The cycle time
   - **C** ☐ The time to drive from the detector to the stop line
   - **D** ☐ The length of the green

5. **With T = 10 s and β = 0.8, when do the first predicted arrivals reach the stop line?**
   - **A** ☐ After 10 s
   - **B** ☐ After 12.5 s
   - **C** ☐ After 8 s
   - **D** ☐ After 2 s

6. **With α = 0.35, β = 0.8 and T = 10 s, what is the smoothing factor F = 1 ÷ (1 + αβT)?**
   - **A** ☐ 0.50
   - **B** ☐ 0.80
   - **C** ☐ 0.10
   - **D** ☐ 0.263

7. **What does F = 0.263 mean?**
   - **A** ☐ 26% of the cycle is green
   - **B** ☐ The platoon arrives 26% faster
   - **C** ☐ 26% of the cars are lost
   - **D** ☐ 26% of each second's traffic arrives first; 74% spills into later seconds

8. **On a longer road with T = 20 s, F becomes...**
   - **A** ☐ Zero
   - **B** ☐ Larger, so the platoon spreads less
   - **C** ☐ Smaller (about 0.15), so the platoon spreads more
   - **D** ☐ Exactly the same

9. **A steady 1 LPU/s arrives at the loop. Second 1 at the stop line is 0.263. What is second 2?**
   - **A** ☐ 0.526
   - **B** ☐ 0.457
   - **C** ☐ 0.263
   - **D** ☐ 0.737

10. **Compared with the platoon at the detector, the platoon at the stop line has...**
    - **A** ☐ Fewer vehicles
    - **B** ☐ Exactly the same shape
    - **C** ☐ The same vehicles, a lower peak and a longer duration
    - **D** ☐ More vehicles

GROUP 3 · PLATOONS SPREAD · QUIZ 2 OF 6

Key words

Names: _____ Score: ____ / 10

1. **What does dispersion mean in SCOOT's link model?**
   - **A** ☐ The detector missing some vehicles
   - **B** ☐ The platoon spreading out between the detector and the stop line
   - **C** ☐ The queue draining during green
   - **D** ☐ Cars leaving the road at side turnings

2. **Which model does SCOOT use to spread the platoon on its way to the stop line?**
   - **A** ☐ The degree-of-saturation rule
   - **B** ☐ The bathtub queue model
   - **C** ☐ Robertson's dispersion model
   - **D** ☐ Webster's optimum cycle formula

3. **What is TRANSYT, in this course?**
   - **A** ☐ The name of SCOOT's split optimiser
   - **B** ☐ A type of bus priority lane
   - **C** ☐ The detector loop SCOOT reads 4 times a second
   - **D** ☐ An off-line design program that uses the same profile and dispersion model as SCOOT

4. **In $F = 1 \div (1 + \alpha\beta T)$, what is the smoothing factor F?**
   - **A** ☐ The speed of the platoon in m/s
   - **B** ☐ The fraction of the cycle that is green
   - **C** ☐ The share of each second's detected traffic that arrives in the matching second at the stop line
   - **D** ☐ The share of cars that are lost on the link

5. **What are α and β in Robertson's model?**
   - **A** ☐ The smoothing weights 0.75 and 0.25
   - **B** ☐ Dispersion constants, 0.35 and 0.8, that engineers can tune
   - **C** ☐ The cycle time and the green time
   - **D** ☐ The start lag and end lag, 3 s and 4 s

6. **What is the lag βT?**
   - **A** ☐ The time from detection until the first predicted arrivals reach the stop line
   - **B** ☐ The time between two stage changes
   - **C** ☐ The time for the queue to clear
   - **D** ☐ The time the loop is covered by one car

7. **In SCOOT words, what is a link?**
   - **A** ☐ One approach road, from its detector to its stop line
   - **B** ☐ One stage of the signal plan
   - **C** ☐ The cable joining two traffic lights
   - **D** ☐ A group of junctions sharing a cycle

8. **The cyclic flow profile is drawn against "second of the cycle". Why that axis?**
   - **A** ☐ It shows how long each car takes to cross the loop
   - **B** ☐ It counts the cycles since the system started
   - **C** ☐ It shows the time of day of the rush hour
   - **D** ☐ The signals repeat every cycle, so it shows where in the cycle platoons arrive

9. **What does profile smoothing (new = 0.75 × old + 0.25 × this cycle) do?**
   - **A** ☐ Spreads the platoon from detector to stop line
   - **B** ☐ Shortens the cycle by 25%
   - **C** ☐ Blends each new cycle in gently so one odd cycle cannot dominate
   - **D** ☐ Removes buses from the count

10. **In the stop-line curve of a dispersed platoon, what is the tail?**

A ☐ The slower drivers who arrive after the main bunch

B ☐ The cars waiting in the queue from the last cycle

C ☐ The first cars to arrive

D ☐ The cars lost on the way

GROUP 3 · PLATOONS SPREAD · QUIZ 3 OF 6

Numbers and calculations

Names: _____ Score: ____ / 10

1. **A link is 180 m long and cars cruise at 12 m/s. What are the cruise time T and the lag βT?**
   - **A** ☐ T = 15 s, lag 15 s
   - **B** ☐ T = 0.067 s, lag 0.053 s
   - **C** ☐ T = 12 s, lag 9.6 s
   - **D** ☐ T = 15 s, lag 12 s

2. **Find F for a short link with cruise time T = 5 s.**
   - **A** ☐ 0.583
   - **B** ☐ 0.417
   - **C** ☐ 0.364
   - **D** ☐ 0.200

3. **Find F for a long link with cruise time T = 25 s.**
   - **A** ☐ 0.143
   - **B** ☐ 0.875
   - **C** ☐ 0.103
   - **D** ☐ 0.125

4. **The course's long-link example has T = 20 s. What is the lag?**
   - **A** ☐ 25 s
   - **B** ☐ 4 s
   - **C** ☐ 16 s
   - **D** ☐ 20 s

5. **Worked example (T = 10 s, F = 0.263, steady 1 LPU/s at the loop): second 2 at the stop line was 0.457. What is second 3?**
   - **A** ☐ 0.600
   - **B** ☐ 0.337
   - **C** ☐ 0.720
   - **D** ☐ 0.457

6. **In the worked example the stop-line arrival peaks at 0.95, then the platoon at the loop ends (0 LPU/s). What is the next second's arrival (F = 0.263)?**
   - **A** ☐ 0.70
   - **B** ☐ 0.95
   - **C** ☐ 0
   - **D** ☐ 0.25

7. **A profile bar was 4 LPU. This cycle the loop saw 8 LPU in that second. With smoothing new = 0.75 × old + 0.25 × new, what is the bar now?**
   - **A** ☐ 8.0 LPU
   - **B** ☐ 7.0 LPU
   - **C** ☐ 5.0 LPU
   - **D** ☐ 6.0 LPU

8. **Suppose F = 0.5. This second the loop sees 2 LPU and last second's stop-line arrival was 1 LPU. What is this second's arrival?**
   - **A** ☐ 3.0 LPU
   - **B** ☐ 1.5 LPU
   - **C** ☐ 1.0 LPU
   - **D** ☐ 2.0 LPU

9. **A platoon crosses the loop at 2 LPU/s for 6 s. How many LPU in total reach the stop line after dispersion?**
   - **A** ☐ 6 LPU
   - **B** ☐ Fewer than 12, because the peak is lower
   - **C** ☐ 12 LPU
   - **D** ☐ 24 LPU

10. **On the course's 10-second link (F = 0.263), the first second of a platoon brings 2 LPU at the loop. What is the first predicted second at the stop line?**
    - **A** ☐ 2.0 LPU
    - **B** ☐ 0.526 LPU
    - **C** ☐ 1.474 LPU
    - **D** ☐ 0.263 LPU

GROUP 3 · PLATOONS SPREAD · QUIZ 4 OF 6

What happens if…?

Names: _____ Score: ____ / 10

1. **A link is shortened so its cruise time falls from 20 s to 10 s. What happens to F and the platoon?**
   - **A** ☐ F falls and the platoon spreads more
   - **B** ☐ F rises (0.152 → 0.263) and the platoon stays tighter
   - **C** ☐ F stays the same; only the lag changes
   - **D** ☐ The platoon loses fewer cars

2. **Cars drive the same link more slowly, so the cruise time grows. What happens to the lag βT?**
   - **A** ☐ It stays at 8 s on every link
   - **B** ☐ It becomes equal to the cycle time
   - **C** ☐ It grows, so the first arrivals come later
   - **D** ☐ It shrinks, so the first arrivals come sooner

3. **An engineer tunes α upwards for a road where drivers' speeds vary a lot. What happens?**
   - **A** ☐ F rises and the platoon is predicted tighter
   - **B** ☐ More vehicles are predicted to arrive
   - **C** ☐ αβT grows, F falls and the model spreads the platoon more
   - **D** ☐ The lag becomes zero

4. **The platoon at the loop suddenly ends. What happens to the predicted stop-line arrivals?**
   - **A** ☐ They drop to zero in the same second
   - **B** ☐ They stay at the peak value for the rest of the cycle
   - **C** ☐ They fade out gradually, keeping 74% of the previous second each second (T = 10 s)
   - **D** ☐ They jump up to a new peak

5. **The platoon seen at the loop doubles from 1 to 2 LPU/s, everything else equal. What happens to the stop-line curve?**
   - **A** ☐ Only the first second doubles
   - **B** ☐ It spreads over twice as long
   - **C** ☐ It arrives twice as early
   - **D** ☐ Every value doubles; the shape and timing stay the same

6. **One cycle a bus breakdown upstream gives an unusually empty second in the profile. What happens to that profile bar?**
   - **A** ☐ Nothing; the profile never changes
   - **B** ☐ It is replaced completely by the odd value
   - **C** ☐ It moves only a quarter of the way towards the odd value
   - **D** ☐ It is deleted from the profile

7. **If SCOOT used the detector profile at the stop line without shifting it forward, what would go wrong?**
   - **A** ☐ It would expect the cars too early, roughly 0.8 T before they really arrive
   - **B** ☐ It would count fewer cars
   - **C** ☐ Nothing; the stop line sees the same profile
   - **D** ☐ It would expect the cars too late

8. **Two junctions are far apart, so the platoon has smeared almost into steady flow. What happens to the benefit of coordinating them?**
   - **A** ☐ It becomes smaller, because there is no tight bunch left to catch on green
   - **B** ☐ It becomes larger, because the platoon is longer
   - **C** ☐ It is unchanged by distance
   - **D** ☐ Coordination becomes impossible because cars are lost

9. **Imagine a link so short that T is almost 0. What does Robertson's recipe predict?**
   - **A** ☐ F ≈ 0, so the platoon spreads forever
   - **B** ☐ F ≈ 1, so the stop-line arrivals copy the detector profile with almost no spreading
   - **C** ☐ Half the cars disappear
   - **D** ☐ The lag becomes very long

10. **Dispersion lowers and lengthens the platoon. What happens to the total LPU arriving in the cycle?**

A ☐ It falls by the share F

B ☐ It rises because the platoon is longer

C ☐ It stays the same

D ☐ It falls to about 0.95 of the original

GROUP 3 · PLATOONS SPREAD · QUIZ 5 OF 6

Everyday pictures

Names: _____ Score: ____ / 10

1. **In the course's picture of a school class walking from the gate to the bus stop, what stands for dispersion?**

   A ☐ Fast walkers pulling ahead and slow ones falling behind

   B ☐ The teacher counting the children at the gate

   C ☐ Children getting lost on the way

   D ☐ The bus arriving late

2. **In the school-class picture, what does the bus stop stand for?**

   A ☐ The upstream detector

   B ☐ The traffic control centre

   C ☐ The cycle time

   D ☐ The stop line

3. **"At the bus stop the class arrives over a longer time, and the busiest moment is less busy." Which result does this picture?**

   A ☐ The platoon arriving exactly T seconds later, unchanged

   B ☐ More vehicles at the stop line

   C ☐ Same vehicles, a lower peak and a longer duration

   D ☐ A higher peak at the stop line

4. **Where does the school-class analogy break down, according to the course?**

   A ☐ Cars cannot overtake freely on a busy road, and the model's constants are tuned

   B ☐ Cars never travel at different speeds

   C ☐ Children are counted in LPUs

   D ☐ Buses never stop at bus stops

5. **In the memory palace, a marching band of 35 players crosses a bridge under a sign "walk at 80% speed". What do 35 and 80% encode?**

   A ☐ 35 m of road at 80 km/h

   B ☐ A 35 s green and an 80 s cycle

   C ☐ $\alpha = 0.35$ and $\beta = 0.8$

   D ☐ 35 LPU and 80% DoS

6. **In the marching-band scene, what does the far end of the bridge stand for?**

   A ☐ The start of the green

   B ☐ The region's cycle time

   C ☐ The stop line, where the spread-out band arrives

   D ☐ The upstream detector

7. **In the platoon-spreading animation, green dots are fast drivers and orange dots are slow ones. What should you notice at the stop line compared with the loop?**

   A ☐ The orange dots have disappeared

   B ☐ The same cars arrive over a longer time with a lower peak

   C ☐ There are more green dots than at the loop

   D ☐ The cars arrive in a tighter bunch

8. **In the "Robertson's recipe" animation, what do the dark and light parts of each bar show?**

   A ☐ Dark = green time; light = red time

   B ☐ Dark = the queue; light = the cars that left

   C ☐ Dark = F × traffic detected now; light = (1 − F) × last second's arrival carried over

   D ☐ Dark = cars; light = buses

9. **The course chart shows the platoon "at the detector" and "at the stop line" (T = 10 s). What is true about the two curves?**

   A ☐ The stop-line curve starts about 8 s later, rises gradually, and has the same area (10 LPU)

   B ☐ Both curves start at the same moment

   C ☐ The stop-line curve is taller and narrower

   D ☐ The stop-line curve has a smaller area because cars are lost

10. **The cyclic-flow-profile animation runs five cycles with two charts. What do they show?**

A ☐ Top: the queue; bottom: the green times

B ☐ Top: the main road; bottom: the side road

C ☐ Top: cars; bottom: buses

D ☐ Top: this cycle's counts; bottom: the smoothed profile SCOOT keeps

GROUP 3 · PLATOONS SPREAD · QUIZ 6 OF 6

Exam challenge

Names: _____ Score: ____ / 10

1. **Worked example (T = 10 s, steady 1 LPU/s): the first three stop-line seconds are 0.263, 0.457 and 0.600. How many LPU have arrived after 3 s?**

   A ☐ 0.60 LPU
   B ☐ 1.32 LPU
   C ☐ 3.00 LPU
   D ☐ 0.79 LPU

2. **Junctions are 300 m apart and cars cruise at 15 m/s. Using Robertson's model, find T, the lag and F.**

   A ☐ T = 20 s, lag 16 s, F = 0.848
   B ☐ T = 20 s, lag 20 s, F = 0.263
   C ☐ T = 20 s, lag 16 s, F = 0.152
   D ☐ T = 4500 s, lag 3600 s, F ≈ 0

3. **A student says: "On a 10 s link the platoon's front reaches the stop line 10 s after the loop." What is the best correction?**

   A ☐ The front arrives at βT = 8 s, because the fastest drivers lead; others follow in the tail
   B ☐ The front arrives at 2.63 s, because F = 0.263
   C ☐ The front arrives at once, because the loop is at the stop line
   D ☐ The front arrives at 12.5 s, because cars slow down

4. **Which of these four links spreads a platoon the most?**

   A ☐ Cruise time 10 s (F = 0.263)
   B ☐ Cruise time 5 s (F = 0.417)
   C ☐ Cruise time 15 s (F = 0.192)
   D ☐ Cruise time 25 s (F = 0.125)

5. **Why does the worked-example peak reach only 0.95 and not 1.0 LPU/s?**

   A ☐ The detector misses one car in twenty
   B ☐ The platoon at the loop ends after 10 s, before the running average catches up with 1.0
   C ☐ β = 0.8 removes 20% of the traffic
   D ☐ 5% of the cars are lost on the link

6. **A profile bar is 4 LPU. Two cycles in a row each see 8 LPU in that second. Using new = 0.75 × old + 0.25 × new, what is the bar after the second cycle?**

   A ☐ 8.0 LPU
   B ☐ 5.0 LPU
   C ☐ 6.0 LPU
   D ☐ 5.75 LPU

7. **Dispersion changes a link's predicted arrivals. Which does it change?**

   A ☐ The total LPU arriving per cycle, which sets the DoS
   B ☐ The number of vehicles counted at the loop
   C ☐ The saturation occupancy SATO
   D ☐ When in the cycle the cars arrive, which matters for the best offset

8. **Neighbouring group (offsets): junctions are 10 s of cruise time apart. Why should SCOOT use the dispersed stop-line profile, not the loop profile, when judging offsets?**

   A ☐ The green should meet the cars when they really reach the stop line, which is later and more spread than at the loop
   B ☐ The loop profile has fewer cars in it
   C ☐ Offsets are chosen only from the cycle time
   D ☐ The loop profile already shows the queue

9. **The same platoon is detected on two links: A with T = 10 s and B with T = 20 s. Which statement is right?**

   A ☐ B: the same lag but more cars
   B ☐ B: first arrivals later (16 s vs 8 s) and a flatter peak (F 0.152 vs 0.263)
   C ☐ B: first arrivals earlier and a taller peak
   D ☐ A and B give identical stop-line curves

10. **Neighbouring group (queue model): what role do Robertson's predicted stop-line arrivals play in the bathtub picture?**

A ☐ They are the tap: the arrival rate filling the queue during red

B ☐ They are the plughole draining the queue

C ☐ They are the plug, deciding when green starts

D ☐ They are the size of the bathtub

GROUP 4 · QUEUE AND DEGREE OF SATURATION · QUIZ 1 OF 6

Basics

Names: _____ Score: ____ / 10

1. **In the bathtub picture of a queue, what is the tap?**
   - **A** ☐ The saturation occupancy
   - **B** ☐ The arriving traffic
   - **C** ☐ The queue
   - **D** ☐ The green light

2. **What is SATO?**
   - **A** ☐ The amber time
   - **B** ☐ The average speed on the link
   - **C** ☐ Saturation occupancy: how fast a standing queue drains in green
   - **D** ☐ The offset between junctions

3. **Arrivals are 0.2 LPU/s and red lasts 35 s. How big is the queue at the end of red?**
   - **A** ☐ 5 LPU
   - **B** ☐ 3.5 LPU
   - **C** ☐ 12 LPU
   - **D** ☐ 7 LPU

4. **In green, SATO is 0.6 LPU/s and cars still arrive at 0.2 LPU/s. How fast does the queue shrink?**
   - **A** ☐ 0.6 LPU/s
   - **B** ☐ 0.8 LPU/s
   - **C** ☐ 0.2 LPU/s
   - **D** ☐ 0.4 LPU/s

5. **How long does a 7 LPU queue take to clear at 0.4 LPU/s?**
   - **A** ☐ 28 s
   - **B** ☐ 17.5 s
   - **C** ☐ 7 s
   - **D** ☐ 12 s

6. **What is the degree of saturation (DoS)?**
   - **A** ☐ Delay ÷ stops
   - **B** ☐ Demand ÷ what the green can clear
   - **C** ☐ Green time ÷ cycle time
   - **D** ☐ Queue × SATO

7. **Demand is 0.2 × 60 = 12 LPU per cycle and the green can clear 25 × 0.6 = 15 LPU. What is the DoS?**
   - **A** ☐ 1.25
   - **B** ☐ 0.80
   - **C** ☐ 1.50
   - **D** ☐ 0.60

8. **Why does SCOOT aim for about 90% rather than 100%?**
   - **A** ☐ To save fuel
   - **B** ☐ To leave a margin for random surges in traffic
   - **C** ☐ Because detectors fail above 90%
   - **D** ☐ Because 100% is illegal

9. **A link has a DoS of 1.10. What happens?**
   - **A** ☐ The queue grows every cycle
   - **B** ☐ There is plenty of spare green
   - **C** ☐ There is no traffic
   - **D** ☐ The green is exactly used up

10. **In the queue model, total delay equals...**
    - **A** ☐ The number of stops
    - **B** ☐ SATO × green time
    - **C** ☐ The area under the queue-over-time line
    - **D** ☐ The length of the red

GROUP 4 · QUEUE AND DEGREE OF SATURATION · QUIZ 2 OF 6

Key words

Names: _____ Score: ____ / 10

1. **What is effective green?**

   A ☐ The green shown on the signal head, exactly as displayed

   B ☐ The green given to the busiest stage only

   C ☐ The part of the cycle when the queue really discharges: it starts a start lag after green and runs an end lag into amber

   D ☐ The green time minus the whole intergreen

2. **What is the start lag?**

   A ☐ The few seconds after the light turns green before the queue starts moving at full rate, because drivers react

   B ☐ The delay before SCOOT changes the cycle time

   C ☐ The amber time at the end of green

   D ☐ The time a car takes from the detector to the stop line

3. **What is the end lag?**

   A ☐ The time the last car waits in the queue

   B ☐ The gap between two cycles

   C ☐ The extra time after the light turns amber during which drivers keep going through

   D ☐ The time the detector takes to send its count

4. **What is a residual queue?**

   A ☐ Queue left over at the end of green that carries into the next cycle

   B ☐ The queue that the detector measures directly

   C ☐ The queue at the end of red

   D ☐ The cars waiting on a side road

5. **In the queue model, which vehicles count as stops?**

   A ☐ Only the first car in the queue

   B ☐ Only vehicles that arrive in green

   C ☐ Every vehicle that crosses the detector

   D ☐ Arrivals that meet a red light or a queue

6. **What does the performance index (PI) measure?**

   A ☐ The degree of saturation of the worst link

   B ☐ How long the cycle is compared with Webster's cycle

   C ☐ The number of LPUs counted per hour

   D ☐ Delay plus K times stops: the score the optimisers try to make as low as possible

7. **In PI = delay + K × stops, what is K?**

   A ☐ The saturation occupancy

   B ☐ A weight that sets how much one stop matters compared with delay

   C ☐ The cruise time of the link

   D ☐ The number of lanes

8. **The model's delay is given in which unit?**

   A ☐ Vehicles per hour

   B ☐ LPU·seconds

   C ☐ LPU per second

   D ☐ Seconds per cycle

9. **What is the saturation flow s of a lane?**

   A ☐ The top rate a standing queue leaves in green, about 1800 veh/h per lane

   B ☐ The average flow over a whole day

   C ☐ The flow at which the detector stops working

   D ☐ The flow of the first car after the light turns green

10. **What does the capacity of an approach mean?**

- **A** ☐ The green time divided by the red time
- **B** ☐ The arrival rate in the peak hour
- **C** ☐ What the approach can carry per hour, s × g ÷ C
- **D** ☐ The longest queue the road can hold

GROUP 4 · QUEUE AND DEGREE OF SATURATION · QUIZ 3 OF 6

Numbers and calculations

Names: _____ Score: ____ / 10

1. **Arrivals are 0.25 LPU/s and effective red lasts 40 s, with no queue at the start. How big is the queue at the end of red?**

   A ☐ 15 LPU **B** ☐ 2.5 LPU
   C ☐ 10 LPU **D** ☐ 160 LPU

2. **A 10 LPU queue faces green with SATO 0.75 LPU/s while traffic still arrives at 0.25 LPU/s. How long until it clears?**

   A ☐ 10 s **B** ☐ 13.3 s
   C ☐ 20 s **D** ☐ 40 s

3. **Arrivals 0.15 LPU/s, cycle 80 s, effective green 40 s, SATO 0.5 LPU/s, no residual. What is the DoS?**

   A ☐ 0.60 **B** ☐ 1.67
   C ☐ 0.50 **D** ☐ 0.30

4. **This cycle 15 LPU arrive and 3 LPU were left over from last cycle. Effective green 25 s, SATO 0.8 LPU/s. What is the DoS?**

   A ☐ 0.75 **B** ☐ 0.90
   C ☐ 1.11 **D** ☐ 0.72

5. **Displayed green is 30 s, start lag 3 s, end lag 2 s. What is the effective green?**

   A ☐ 29 s **B** ☐ 31 s
   C ☐ 35 s **D** ☐ 25 s

6. **Arrivals 0.2 LPU/s, red 30 s, SATO 0.5 LPU/s, no starting queue. The queue is 6 LPU at the end of red and clears 20 s into green. What is the delay per cycle?**

   A ☐ 300 LPU·s **B** ☐ 60 LPU·s
   C ☐ 150 LPU·s **D** ☐ 90 LPU·s

7. **An option gives delay 150 LPU·s and 10 LPU of stops, with K = 10. What is its PI?**

   A ☐ 100 **B** ☐ 1600
   C ☐ 250 **D** ☐ 160

8. **Course example: arrivals 0.2 LPU/s, red 35 s, queue clears 17.5 s into green. How many LPU are stopped per cycle?**

   A ☐ 10.5 LPU **B** ☐ 12 LPU
   C ☐ 7 LPU **D** ☐ 3.5 LPU

9. **Course example: s = 1800 veh/h, g = 25 s, C = 60 s, demand 540 veh/h. What is the DoS?**

   A ☐ 0.42 **B** ☐ 1.39
   C ☐ 0.30 **D** ☐ 0.72

10. **Course DoS animation: demand 14.4 LPU, capacity 15 LPU (25 s × 0.6). The split optimiser adds 4 s of green. New DoS?**

    A ☐ 0.96 **B** ☐ 0.76
    C ☐ 0.83 **D** ☐ 0.80

GROUP 4 · QUEUE AND DEGREE OF SATURATION · QUIZ 4 OF 6

What happens if…?

Names: _____ Score: ____ / 10

1. **Arrivals on a link rise but its green stays the same. What happens to the DoS?**
   - **A** ☐ It falls
   - **B** ☐ It stays the same
   - **C** ☐ It rises
   - **D** ☐ It drops to zero

2. **The split optimiser gives a stage a few seconds more effective green, with the same arrivals. What happens to that link's DoS?**
   - **A** ☐ It rises, because more cars arrive
   - **B** ☐ It doubles
   - **C** ☐ It falls, because the green can clear more
   - **D** ☐ It stays the same

3. **Two lanes have the same queue, but one has a higher SATO. What happens?**
   - **A** ☐ The higher-SATO lane empties sooner and needs less green
   - **B** ☐ The higher-SATO lane needs more green
   - **C** ☐ Both empty at the same time
   - **D** ☐ The higher-SATO lane builds a longer queue in red

4. **A residual queue is left at the end of green. What happens to next cycle's DoS, other things equal?**
   - **A** ☐ It goes down, because fewer cars arrive
   - **B** ☐ It resets to zero
   - **C** ☐ It is unchanged; residuals are ignored
   - **D** ☐ It goes up, because the leftover queue is added to the demand

5. **Drivers react more slowly, so the start lag gets longer. What happens?**
   - **A** ☐ Nothing; DoS uses the displayed green
   - **B** ☐ Only the red time changes
   - **C** ☐ Effective green gets shorter and the DoS rises
   - **D** ☐ Effective green gets longer and the DoS falls

6. **During green, cars arrive faster than SATO. What happens to the queue in green?**
   - **A** ☐ It stays exactly the same size
   - **B** ☐ It keeps growing instead of shrinking
   - **C** ☐ It clears at once
   - **D** ☐ It shrinks faster than usual

7. **The red is made longer while arrivals stay the same (and green can still clear the queue). What happens to delay per cycle?**
   - **A** ☐ It falls
   - **B** ☐ It grows, because the queue gets taller and takes longer to clear
   - **C** ☐ It becomes zero
   - **D** ☐ It stays the same

8. **The weight K in the PI is raised. What happens when the optimisers compare options?**
   - **A** ☐ Stops no longer count
   - **B** ☐ The DoS target rises to 100%
   - **C** ☐ Options that cause more stops look worse than before
   - **D** ☐ Delay no longer counts

9. **The model's SATO is set higher than the real street's. What happens to the modelled queue?**
   - **A** ☐ It clears at exactly the same time
   - **B** ☐ It never forms
   - **C** ☐ It clears earlier than the real queue
   - **D** ☐ It clears later than the real queue

10. **A link has DoS 0.80, under the 90% target. What does this allow SCOOT to do?**
    - **A** ☐ Nothing; DoS under 90% is a fault
    - **B** ☐ Give some of its spare green to a busier stage, a few seconds at a time
    - **C** ☐ Lengthen this link's green a lot
    - **D** ☐ Switch the junction off

GROUP 4 · QUEUE AND DEGREE OF SATURATION · QUIZ 5 OF 6

Everyday pictures

Names: _____ Score: ____ / 10

1. **In the bathtub picture of a queue, what is the plug being in?**
   - **A** ☐ The detector
   - **B** ☐ The green light
   - **C** ☐ The red light
   - **D** ☐ The arriving traffic

2. **In the bathtub picture, what is the water level in the tub?**
   - **A** ☐ The number of lanes
   - **B** ☐ The cruise time
   - **C** ☐ The queue at the stop line
   - **D** ☐ The cycle time

3. **In the bathtub picture, the tap runs slower than the drain. What happens during green?**
   - **A** ☐ The tub empties
   - **B** ☐ The tub overflows
   - **C** ☐ The plug goes back in
   - **D** ☐ The water level stays the same

4. **Where does the simple bathtub picture break down?**
   - **A** ☐ A real queue has a length on the road and can block the junction behind it
   - **B** ☐ Real traffic never arrives during green
   - **C** ☐ Real queues do not grow in red
   - **D** ☐ Real queues drain at no fixed rate at all

5. **In the memory palace the bathtub has a gauge painted red above one mark. What does that mark stand for?**
   - **A** ☐ SCOOT's 90% DoS target
   - **B** ☐ The 10–20 s warning
   - **C** ☐ The 60 s cycle
   - **D** ☐ The SATO value

6. **SCOOT nudges a link's green as its DoS drifts, like a thermostat. In that picture, what plays the room temperature?**
   - **A** ☐ The modelled DoS of the link
   - **B** ☐ The detector cable
   - **C** ☐ The colour of the signal
   - **D** ☐ The number of junctions in the region

7. **In the course's queue chart (red 0–35 s, green 35–60 s), why does the line fall more steeply than it rose?**
   - **A** ☐ Cars stop arriving in green
   - **B** ☐ The detector counts twice in green
   - **C** ☐ It drains at 0.4 LPU/s but filled at only 0.2 LPU/s
   - **D** ☐ The cycle gets shorter in green

8. **In the same queue chart, how long is the queue empty at the end of green?**
   - **A** ☐ 0 s
   - **B** ☐ 7.5 s
   - **C** ☐ 25 s
   - **D** ☐ 17.5 s

9. **In the degree-of-saturation animation the gauge moves into the red. What brings it back?**
   - **A** ☐ The detector is switched off
   - **B** ☐ The split optimiser gives the stage about 4 s more green
   - **C** ☐ The cycle is cut in half
   - **D** ☐ The arrivals are ignored

10. **In the SATO animation two lanes hold the same six-car queue, one with SATO 0.6 LPU/s and one with 1.0 LPU/s. What do you see?**
    - **A** ☐ The 0.6 LPU/s lane empties sooner
    - **B** ☐ Both empty together
    - **C** ☐ Neither queue moves
    - **D** ☐ The 1.0 LPU/s lane empties sooner

GROUP 4 · QUEUE AND DEGREE OF SATURATION · QUIZ 6 OF 6

Exam challenge

Names: _____ Score: ____ / 10

1. **Displayed green 27 s, start lag 3 s, end lag 1 s, SATO 0.6 LPU/s, demand 12 LPU per cycle, no residual. What is the DoS?**

   A ☐ 0.80 **B** ☐ 0.74
   C ☐ 1.35 **D** ☐ 0.83

2. **Demand is 12 LPU per cycle and SATO is 0.6 LPU/s. What is the shortest effective green that keeps the DoS at 90%?**

   A ☐ About 22.2 s **B** ☐ 25 s
   C ☐ 20 s **D** ☐ 18 s

3. **Every cycle 18 LPU arrive but the green can clear only 15 LPU. Starting with no queue, how big is the residual after 3 cycles?**

   A ☐ 3 LPU **B** ☐ 54 LPU
   C ☐ 9 LPU **D** ☐ 6 LPU

4. **A link sits at DoS exactly 1.00 every cycle. Which statement is right?**

   A ☐ The road is jammed forever **B** ☐ There is plenty of spare green
   C ☐ The queue grows every cycle **D** ☐ The green is just used up; a random surge would leave a residual queue

5. **K = 10. Option A: delay 200 LPU·s, 12 LPU stops. Option B: delay 180 LPU·s, 15 LPU stops. Which does the optimiser prefer?**

   A ☐ Neither; they are equal **B** ☐ B, with PI 195 against 212
   C ☐ B, because its delay is lower **D** ☐ A, with PI 320 against 330

6. **Linking to validation: an engineer sees an 18 LPU queue clear at 33 s with a 3 s start lag. What SATO should the model use?**

   A ☐ 1.67 LPU/s **B** ☐ 0.55 LPU/s
   C ☐ 0.67 LPU/s **D** ☐ 0.60 LPU/s

7. **s = 1800 veh/h, g = 30 s, C = 90 s, demand 480 veh/h. Capacity and DoS?**

   A ☐ 1800 veh/h and 0.27 **B** ☐ 600 veh/h and 0.80
   C ☐ 900 veh/h and 0.53 **D** ☐ 600 veh/h and 1.25

8. **In the course example (0.2 LPU/s, SATO 0.6) the delay is 183.75 LPU·s. If the red were twice as long (70 s) and green could still clear the queue, the delay would be...**

   A ☐ Twice as big (367.5 LPU·s) **B** ☐ About four times as big (735 LPU·s)
   C ☐ Half as big **D** ☐ The same

9. **Linking to the cycle optimiser: splits are already balanced but the critical junction stays above about 90%. What does SCOOT do over the next minutes?**

   A ☐ Shortens every green by 4 s **B** ☐ Changes the offset once a second
   C ☐ Switches to a fixed plan **D** ☐ Lengthens the cycle, checking every 5 minutes (2.5 when rising)

10. **Last cycle left 1.5 LPU behind. This cycle 12 LPU arrive and the green can clear 15 LPU. What is the DoS, and how does it compare with the target?**

A ☐ 0.70: plenty of spare green

B ☐ 0.90: right at SCOOT's target

C ☐ 0.80: well under the target

D ☐ 1.10: overloaded

GROUP 5 · VALIDATION · QUIZ 1 OF 6

Basics

Names: _____ Score: ____ / 10

1. **What is validation in SCOOT?**

   A ☐ Tuning each link's model so it matches what happens on the street

   B ☐ Checking the signal bulbs

   C ☐ Counting accidents

   D ☐ Installing new loops

2. **The model and the engineer's stopwatch disagree about when the queue clears. Which is right?**

   A ☐ The average of the two

   B ☐ The street (the stopwatch)

   C ☐ Neither

   D ☐ The model

3. **Which formula gives SATO from a stopwatch measurement?**

   A ☐ SATO = (clear time + start lag) ÷ queue

   B ☐ SATO = queue × start lag

   C ☐ SATO = clear time ÷ queue

   D ☐ SATO = queue ÷ (clear time − start lag)

4. **A 19.5 LPU queue clears 36 s after green starts; the start lag is 3 s. What is SATO?**

   A ☐ 0.81 LPU/s

   B ☐ 0.67 LPU/s

   C ☐ 0.54 LPU/s

   D ☐ 0.59 LPU/s

5. **With SATO set to 0.67 LPU/s, when does the model say that 19.5 LPU queue clears (start lag 3 s)?**

   A ☐ After 39 s

   B ☐ After about 32 s

   C ☐ After 36 s

   D ☐ After about 29 s

6. **What goes wrong if SATO is set too high?**

   A ☐ SCOOT stops working

   B ☐ The cycle doubles

   C ☐ The model thinks queues clear sooner, gives too little green, and real queues grow

   D ☐ SCOOT gives far too much green

7. **What is the start lag?**

   A ☐ The time drivers take to get going after the green starts

   B ☐ The length of the cycle

   C ☐ The amber time

   D ☐ The travel time between junctions

8. **Which of these is NOT a setting checked during validation?**

   A ☐ Journey (cruise) time

   B ☐ The colour of the signal heads

   C ☐ Saturation occupancy (SATO)

   D ☐ Start and end lags

9. **Which street features can lower SATO?**

   A ☐ Freshly painted markings

   B ☐ Wide straight lanes

   C ☐ Extra detectors

   D ☐ Narrow lanes, uphill gradients, parking and turning traffic

10. **Which analogy fits validation best?**

    A ☐ Buying a new car

    B ☐ Reading a map

    C ☐ Tuning a guitar by ear until it matches a tuning fork

    D ☐ Painting a road

GROUP 5 · VALIDATION · QUIZ 2 OF 6

Key words

Names: _____ Score: ____ / 10

1. **What does SATO (saturation occupancy) describe?**
   - **A** ☐ How many cars arrive at the detector each cycle
   - **B** ☐ The share of the cycle that is green
   - **C** ☐ The time from the detector to the stop line
   - **D** ☐ How fast a standing queue drains across the stop line in green, in LPU per second

2. **What are the units of SATO?**
   - **A** ☐ Percent
   - **B** ☐ Vehicles per hour per cycle
   - **C** ☐ LPU per second
   - **D** ☐ Seconds

3. **What is the cruise time of a link?**
   - **A** ☐ How long a queue takes to clear
   - **B** ☐ The time between two cycles
   - **C** ☐ How long the green lasts
   - **D** ☐ How long a car takes to drive from the detector to the stop line

4. **What is the end lag?**
   - **A** ☐ The time drivers take to get going after the green starts
   - **B** ☐ The extra time traffic keeps crossing the stop line after the green ends, into the amber
   - **C** ☐ The delay before SCOOT changes the cycle
   - **D** ☐ The all-red time between stages

5. **How does effective green differ from displayed green?**
   - **A** ☐ It is the green shown to pedestrians
   - **B** ☐ It includes the whole intergreen
   - **C** ☐ It is always exactly the same as displayed green
   - **D** ☐ It starts a start lag late and runs an end lag into the amber

6. **Validation sets cruise time, SATO, start lag, end lag and one more setting. Which?**
   - **A** ☐ The colour of the detector loops
   - **B** ☐ The optimiser's −4/0/+4 s trial steps
   - **C** ☐ The maximum queue clear time
   - **D** ☐ The target degree of saturation

7. **In the validation formulas, what is the queue clear time?**
   - **A** ☐ The time after green starts until the queue that was standing at the start of green has gone
   - **B** ☐ The time between two detector readings
   - **C** ☐ The time a single car takes to cross the loop
   - **D** ☐ The length of the red period

8. **What is the model queue in SCOOT?**
   - **A** ☐ The queue SCOOT calculates from upstream detector counts, without ever seeing it
   - **B** ☐ A queue photographed by a camera at the stop line
   - **C** ☐ The longest queue ever recorded on the link
   - **D** ☐ A queue of cars waiting for the detector to be repaired

9. **What is commissioning a link?**
   - **A** ☐ Buying the signal heads
   - **B** ☐ Setting a link up before it goes live, when the engineer compares the model with the street
   - **C** ☐ Choosing the cycle time once a year
   - **D** ☐ Paying the contractor who laid the loops

10. **How is SATO related to saturation flow?**

A ☐ It is the saturation flow multiplied by the cycle time

B ☐ It is the arrival rate at the detector

C ☐ It is the same idea (the top discharge rate of a queue) measured in LPU per second

D ☐ They are unrelated quantities

GROUP 5 · VALIDATION · QUIZ 3 OF 6

Numbers and calculations

Names: _____ Score: ____ / 10

1. **The course's 'try it' example: a queue of 18 LPU clears on the street 27 s after green starts; start lag 3 s. What should SATO be?**
   - **A** ☐ 1.33 LPU/s
   - **B** ☐ 0.75 LPU/s
   - **C** ☐ 0.60 LPU/s
   - **D** ☐ 0.67 LPU/s

2. **In the same 'try it' example the model cleared the 18 LPU queue at 30 s (start lag 3 s). What SATO was the model using?**
   - **A** ☐ About 0.67 LPU/s
   - **B** ☐ 0.60 LPU/s
   - **C** ☐ 0.75 LPU/s
   - **D** ☐ 1.5 LPU/s

3. **Lab F2 measured 1.95 LPU per vehicle. How many LPU is a queue of 10 cars?**
   - **A** ☐ 19.5 LPU
   - **B** ☐ 10 LPU
   - **C** ☐ 29.5 LPU
   - **D** ☐ 1.95 LPU

4. **The course's SATO changed from 0.67 to 0.59 LPU/s after validation. Roughly what percentage change is that?**
   - **A** ☐ About 12%
   - **B** ☐ About 88%
   - **C** ☐ About 8%
   - **D** ☐ About 4%

5. **On the Lab F2 side road (SATO 0.89 LPU/s, start lag 3 s), when does a 19.5 LPU queue clear?**
   - **A** ☐ About 22 s
   - **B** ☐ About 25 s after green starts
   - **C** ☐ About 17 s
   - **D** ☐ About 32 s

6. **A queue of 12 LPU waits at red. SATO is 0.6 LPU/s and the start lag is 2 s. When does it clear?**
   - **A** ☐ 22 s after green starts
   - **B** ☐ 20 s
   - **C** ☐ 9.2 s
   - **D** ☐ 7.2 s

7. **A 20 LPU queue clears on the street 28 s after green starts; the start lag is 3 s. What is SATO?**
   - **A** ☐ 0.65 LPU/s
   - **B** ☐ 1.25 LPU/s
   - **C** ☐ 0.8 LPU/s
   - **D** ☐ 0.71 LPU/s

8. **A link is 162 m from detector to stop line and cars cruise at the lab's 13.5 m/s. What is the cruise time?**
   - **A** ☐ 13.5 s
   - **B** ☐ 2187 s
   - **C** ☐ 8.3 s
   - **D** ☐ 12 s

9. **Displayed green is 30 s, with the Lab F2 lags (start 3 s, end 4 s). What is the effective green?**
   - **A** ☐ 37 s
   - **B** ☐ 23 s
   - **C** ☐ 30 s
   - **D** ☐ 31 s

10. **The model uses SATO 0.6 LPU/s for a 15 LPU queue (start lag 3 s) and clears it at 28 s, but the street clears at 33 s. What should SATO be?**
    - **A** ☐ 0.5 LPU/s
    - **B** ☐ 0.6 LPU/s
    - **C** ☐ 0.45 LPU/s
    - **D** ☐ 2.0 LPU/s

GROUP 5 · VALIDATION · QUIZ 4 OF 6

What happens if…?

Names: _____ Score: ____ / 10

1. **The model's queue empties earlier than the real queue. What should the engineer do?**
   - **A** ☐ Turn SATO up
   - **B** ☐ Turn SATO down: the model thinks traffic leaves faster than it does
   - **C** ☐ Leave it: the model is right
   - **D** ☐ Raise the target DoS

2. **The model's queue is still showing after the real queue has gone. What does that mean?**
   - **A** ☐ The cycle time is too short
   - **B** ☐ SATO is too low and should be raised
   - **C** ☐ SATO is too high and should be lowered
   - **D** ☐ The detector is counting twice

3. **If the model's SATO is set too low, what happens to the green that link receives?**
   - **A** ☐ Nothing changes
   - **B** ☐ The link gets too little green
   - **C** ☐ The link is skipped every other cycle
   - **D** ☐ The model thinks queues clear slowly, so it tends to give that link more green than it needs

4. **Parking is added along a link and its lanes get narrower. What should happen?**
   - **A** ☐ The link should be re-validated, because SATO is likely to fall
   - **B** ☐ Nothing: SCOOT parameters never change
   - **C** ☐ SATO should be raised
   - **D** ☐ The cruise time should be set to zero

5. **The start lag on a link is really 5 s, not 3 s. With the same queue and SATO, how does the real clear time compare with the model's?**
   - **A** ☐ It is 2 s later
   - **B** ☐ It is 2 s earlier
   - **C** ☐ It is the same
   - **D** ☐ It is 5 s later

6. **The queue at the start of green doubles, but SATO and start lag stay the same. What happens to the clear time?**
   - **A** ☐ The clear time does not change
   - **B** ☐ The whole clear time, including the start lag, doubles
   - **C** ☐ The drain part (queue ÷ SATO) doubles; the start lag stays the same
   - **D** ☐ The clear time halves

7. **With the same arrivals and effective green, SATO in the model is raised. What happens to the calculated DoS?**
   - **A** ☐ It stays exactly the same
   - **B** ☐ It becomes negative
   - **C** ☐ It falls
   - **D** ☐ It rises

8. **The end lag is longer than the model assumes. What does the model get wrong?**
   - **A** ☐ It over-estimates the cruise time
   - **B** ☐ It makes the cycle too long by exactly the end lag
   - **C** ☐ It under-estimates the effective green, so it thinks less traffic can clear each cycle
   - **D** ☐ It counts too many cars at the detector

9. **The cruise time in the model is shorter than the real drive from detector to stop line. What happens?**
   - **A** ☐ The model expects cars at the stop line before they really arrive, so its queue timing is wrong
   - **B** ☐ Nothing, cruise time is not used by the model
   - **C** ☐ The detector reads fewer times a second
   - **D** ☐ SATO automatically corrects itself

10. **SATO is wrong by about 12% (0.67 instead of 0.59 LPU/s) on the 19.5 LPU queue. What is the effect?**

A ☐ The cycle time doubles

B ☐ The detector stops working

C ☐ The model's clear time is about 4 s out, enough to give big errors in delay and DoS

D ☐ Almost no effect: 12% is too small to matter

GROUP 5 · VALIDATION · QUIZ 5 OF 6

Everyday pictures

Names: _____ Score: ____ / 10

1. **In the guitar-tuning picture of validation, what plays the part of the tuning fork?**
   - **A** ☐ The optimiser
   - **B** ☐ The detector loop
   - **C** ☐ The real street
   - **D** ☐ The SCOOT model

2. **In the guitar picture, what is "turning the peg"?**
   - **A** ☐ Changing the cycle time every second
   - **B** ☐ Moving the detector loop
   - **C** ☐ Changing a model setting such as SATO until the model matches the street
   - **D** ☐ Switching the signal off

3. **Where does the guitar analogy break down?**
   - **A** ☐ Validation uses a tuning fork on the street
   - **B** ☐ A guitar stays in tune for weeks; a link may need re-validating whenever lanes, layout or parking change
   - **C** ☐ A guitar never needs tuning at all
   - **D** ☐ A guitar has no strings

4. **"A perfectly built guitar still plays wrong notes if nobody tunes it." What does this say about SCOOT?**
   - **A** ☐ Even a well-designed SCOOT performs poorly if its links are not validated
   - **B** ☐ Only new junctions need SCOOT
   - **C** ☐ SCOOT must be rebuilt every year
   - **D** ☐ SCOOT works best with no detectors

5. **In the memory palace guitar shop, the SATO peg has two small pegs labelled 3 and 4. What do they stand for?**
   - **A** ☐ Start lag 3 s and end lag 4 s
   - **B** ☐ SATO 3 and 4 LPU/s
   - **C** ☐ 3 cars and 4 buses
   - **D** ☐ A 3 s cycle and a 4 s cycle

6. **In the memory palace, the engineer holds a stopwatch in her teeth. What does the stopwatch stand for?**
   - **A** ☐ The cycle optimiser's 5-minute timer
   - **B** ☐ The intergreen
   - **C** ☐ The observed queue clear time on the street
   - **D** ☐ The detector reading 4 times a second

7. **In the bathtub picture of a queue, what does SATO correspond to?**
   - **A** ☐ The size of the tub
   - **B** ☐ How fast the plughole drains the tub when the plug is out (green)
   - **C** ☐ The tap's flow into the tub
   - **D** ☐ The colour of the water

8. **In the SATO animation, the line of cleared LPU starts flat and then rises steadily. What is the slope of the steady part?**
   - **A** ☐ The start lag
   - **B** ☐ The arrival rate at the detector
   - **C** ☐ The cycle time
   - **D** ☐ SATO

9. **The SATO animation shows two lanes with the same six-car queue: SATO 0.6 LPU/s and 1.0 LPU/s. Which lane needs less green?**
   - **A** ☐ The 0.6 LPU/s lane
   - **B** ☐ The 1.0 LPU/s lane: it empties sooner
   - **C** ☐ Neither lane needs any green
   - **D** ☐ Both need exactly the same green

10. **In the Lab F1 figure, the SCOOT model queue follows SUMO's queue but empties a few seconds later. Why?**
    - **A** ☐ SCOOT's detectors are broken
    - **B** ☐ The model sees the real queue with a camera
    - **C** ☐ SUMO uses a different cycle time
    - **D** ☐ SUMO stops counting a car as queued the moment it starts moving

GROUP 5 · VALIDATION · QUIZ 6 OF 6

Exam challenge

Names: _____ Score: ____ / 10

1. **Arrivals plus residual are 18 LPU per cycle and effective green is 30 s. With the old SATO 0.67 the DoS is about 90%. What is it with the validated SATO 0.59 LPU/s?**

   A ☐ About 59%
   B ☐ About 102%: the link is really over capacity
   C ☐ About 79%
   D ☐ About 90%

2. **A colleague says: "The model clears early, so let's lower the target DoS to give more green." What is the best reply?**

   A ☐ Good idea: changing the target is the normal fix
   B ☐ Ignore the stopwatch; the model is right
   C ☐ Fix the model first: correct SATO, because every decision is built on the queue picture
   D ☐ Change the optimiser's trial steps instead

3. **A queue of 10 cars (1.95 LPU each) has SATO 0.65 LPU/s and start lag 3 s. When does it clear?**

   A ☐ 30 s
   B ☐ 33 s after green starts
   C ☐ 18 s
   D ☐ 15.4 s

4. **A 19.5 LPU queue drains on the main road (SATO 0.67) and on the side road (SATO 0.89). How much longer does the drain take on the main road?**

   A ☐ About 29 s
   B ☐ About 22 s
   C ☐ About 0.2 s
   D ☐ About 7 s

5. **Linking to group 3: a link is 135 m long at the lab cruise speed 13.5 m/s. What is the cruise time, and the Robertson F (α = 0.35, β = 0.8)?**

   A ☐ T = 13.5 s, F = 0.8
   B ☐ T = 10 s, F = 0.35
   C ☐ T = 1822 s, F ≈ 0
   D ☐ T = 10 s, F ≈ 0.263

6. **Displayed green is 25 s, lags are 3 s and 4 s, and validated SATO is 0.59 LPU/s. About how many LPU can the green clear?**

   A ☐ About 13.0 LPU
   B ☐ About 15.3 LPU
   C ☐ About 14.8 LPU
   D ☐ About 44 LPU

7. **Why can't one SATO value be copied to every link in a town?**

   A ☐ SCOOT forbids two links having the same number
   B ☐ SATO depends only on the cycle time
   C ☐ SATO depends on lane width, gradient, turning traffic and parking, which differ from street to street
   D ☐ SATO changes every second automatically

8. **Which observation at the stop line shows the model's SATO is too high?**

   A ☐ The real queue clears before the model's
   B ☐ The model shows the queue gone while real cars are still queuing
   C ☐ The signal shows amber
   D ☐ Model and street clear at the same moment

9. **If SATO is left too high, how does the real street suffer?**

   A ☐ The link is given too little green and real queues grow, even though the model looks fine
   B ☐ The detectors stop counting
   C ☐ Drivers get too much green and roads stay empty
   D ☐ The offset optimiser stops running

10. **An engineer measures a 24 LPU queue clearing in 33 s (start lag 3 s), while the model uses 0.67 LPU/s. What is the new SATO, and was the old value too high or too low?**

A ☐ 0.8 LPU/s; the old value was too low

B ☐ 0.8 LPU/s; the old value was too high

C ☐ 1.25 LPU/s; the old value was too high

D ☐ 0.73 LPU/s; the old value was too low

GROUP 6 · SCOOT VS OTHERS · QUIZ 1 OF 6

Basics

Names: _____ Score: ____ / 10

1. **How does a fixed-time signal decide its green?**
   - **A** ☐ It follows a stored plan
   - **B** ☐ It waits for pedestrians
   - **C** ☐ It uses stop-line detectors
   - **D** ☐ It runs a model fed by upstream detectors

2. **How does an actuated signal decide its green?**
   - **A** ☐ It copies its neighbour
   - **B** ☐ It extends the green while cars keep arriving at stop-line loops, and ends it on a gap
   - **C** ☐ It never changes
   - **D** ☐ It runs a model of the whole region

3. **Which description fits SCOOT?**
   - **A** ☐ Upstream loops feed a model; small frequent changes; coordinated junctions
   - **B** ☐ A stored plan only
   - **C** ☐ One loop at the stop line, no model
   - **D** ☐ Random timings

4. **In Lab F3 (one junction in SUMO), which controller had the lowest mean delay?**
   - **A** ☐ Actuated
   - **B** ☐ SCOOT
   - **C** ☐ All the same
   - **D** ☐ Fixed time

5. **SCOOT 12.79 s against fixed time 18.96 s. About how much less delay is that?**
   - **A** ☐ 33%
   - **B** ☐ 12%
   - **C** ☐ 25%
   - **D** ☐ 50%

6. **SCOOT 12.79 s against actuated 14.48 s. About how much less delay?**
   - **A** ☐ 33%
   - **B** ☐ 20%
   - **C** ☐ 5%
   - **D** ☐ 12%

7. **In the four-junction corridor run, which controller had slightly less mean delay?**
   - **A** ☐ Actuated (29.1 s vs SCOOT 30.4 s)
   - **B** ☐ All equal
   - **C** ☐ Fixed time
   - **D** ☐ SCOOT

8. **In that corridor run, what was SCOOT's clearest advantage?**
   - **A** ☐ It used the shortest cycle
   - **B** ☐ It needed no detectors
   - **C** ☐ It was cheapest
   - **D** ☐ Fewer stops and much less bus delay

9. **Where might a simple fixed or actuated controller be good enough?**
   - **A** ☐ Where many buses need priority
   - **B** ☐ On a corridor with closely spaced junctions
   - **C** ☐ In a busy city-centre grid
   - **D** ☐ At an isolated, lightly used junction

10. **What does SCOOT need in order to work well?**
    - **A** ☐ A fixed cycle
    - **B** ☐ GPS in every car
    - **C** ☐ No pedestrians
    - **D** ☐ Working detectors and a validated model

GROUP 6 · SCOOT VS OTHERS · QUIZ 2 OF 6

Key words

Names: _____ Score: ____ / 10

1. **What is TRANSYT?**

   A ☐ An off-line design program for fixed-time network plans

   B ☐ SCOOT's cycle optimiser

   C ☐ A loop placed at the stop line

   D ☐ A type of actuated controller

2. **Why is SCOOT often called an "on-line TRANSYT"?**

   A ☐ It only works where TRANSYT plans already exist

   B ☐ It uses TRANSYT's profile and dispersion model, but runs it live on detector data

   C ☐ It was invented by the TRANSYT team as a joke

   D ☐ It downloads a new TRANSYT plan every hour

3. **In Webster's formula $C_o = (1.5\ L + 5) \div (1 - Y)$, what is L?**

   A ☐ The number of lanes

   B ☐ The lost time per cycle

   C ☐ The loop length

   D ☐ The length of the link

4. **In Webster's formula, what is Y?**

   A ☐ The number of vehicles per cycle

   B ☐ The yellow (amber) time

   C ☐ The target degree of saturation

   D ☐ The sum of the critical flow ratios q/s

5. **What is the flow ratio y of an approach?**

   A ☐ Its saturation flow divided by its flow, s/q

   B ☐ Its green divided by the cycle, g/C

   C ☐ Its delay divided by its stops

   D ☐ Its flow divided by its saturation flow, q/s

6. **What does C_o mean?**

   A ☐ The cycle at time zero

   B ☐ The offset between two junctions

   C ☐ Webster's optimum cycle time for the given flows

   D ☐ The capacity of the busiest lane

7. **In actuated control, what is a gap?**

   A ☐ The all-red time between stages

   B ☐ The space between two junctions

   C ☐ A long enough time between vehicles at the stop-line loop to end the green

   D ☐ A missing detector reading

8. **What is an isolated junction?**

   A ☐ A junction with no traffic

   B ☐ A signalled junction too far from its neighbours to need coordinating with them

   C ☐ A junction that runs only at night

   D ☐ A junction where every loop is broken

9. **What is plan selection?**

   A ☐ SCOOT's small edits to the running plan

   B ☐ Swapping between whole stored plans, an older idea

   C ☐ Choosing where to put the loop

   D ☐ Picking the stage with the highest DoS

10. **Which measure did Lab F3 use to compare the three controllers?**

    A ☐ Cycle length in minutes

    B ☐ Mean delay per vehicle, in seconds

    C ☐ Number of loops used

    D ☐ Fuel price

GROUP 6 · SCOOT VS OTHERS · QUIZ 3 OF 6

Numbers and calculations

Names: _____ Score: ____ / 10

1. **Course example: L = 10 s, y(A) = 0.35, y(B) = 0.25. What is Webster's C_o?**

   A ☐ 50 s **B** ☐ 100 s
   C ☐ 33 s **D** ☐ 40 s

2. **Same example (C_o = 50 s, L = 10 s, Y = 0.60). How much effective green does the main road (y = 0.35) get?**

   A ☐ 29.2 s **B** ☐ 16.7 s
   C ☐ 23.3 s **D** ☐ 17.5 s

3. **Course try: L = 12 s, y(A) = 0.40, y(B) = 0.30. What is C_o?**

   A ☐ 76.7 s **B** ☐ 57.5 s
   C ☐ 32.9 s **D** ☐ 23.0 s

4. **Course trap: L = 10 s and Y = 0.90. What does Webster give for C_o?**

   A ☐ 22 s **B** ☐ 180 s
   C ☐ 50 s **D** ☐ 200 s

5. **Lab F3: fixed time 18.96 s, actuated 14.48 s. About how much less delay did actuation give than fixed time?**

   A ☐ About 12% **B** ☐ About 24%
   C ☐ About 31% **D** ☐ About 4.5%

6. **Corridor run bus delay: actuated 38.0 s, SCOOT 22.9 s. By how much did SCOOT cut bus delay?**

   A ☐ 22.9 s (about 60%) **B** ☐ 15.1 s (about 40%)
   C ☐ 1.3 s (about 4%) **D** ☐ 15.1 s (about 66%)

7. **New numbers: L = 8 s and Y = 0.50. What is C_o?**

   A ☐ 8.5 s **B** ☐ 17 s
   C ☐ 24 s **D** ☐ 34 s

8. **A side road carries 630 veh/h; its saturation flow is 1800 veh/h. What is its flow ratio y?**

   A ☐ 35 **B** ☐ 0.65
   C ☐ 2.86 **D** ☐ 0.35

9. **L = 10 s, y(A) = 0.30, y(B) = 0.20. Webster gives C_o = 40 s. What effective green does the side road B get?**

   A ☐ 12 s **B** ☐ 16 s
   C ☐ 8 s **D** ☐ 18 s

10. **That fixed plan gives road A 18 s of effective green in a 40 s cycle. With s = 1800 veh/h, what is A's capacity?**

    A ☐ 1800 veh/h **B** ☐ 4000 veh/h
    C ☐ 720 veh/h **D** ☐ 810 veh/h

GROUP 6 · SCOOT VS OTHERS · QUIZ 4 OF 6

What happens if…?

Names: _____ Score: ____ / 10

1. **Webster: what happens to C_0 as Y gets close to 1?**
   - **A** ☐ It shrinks to zero
   - **B** ☐ It becomes negative
   - **C** ☐ It stays at 50 s
   - **D** ☐ It grows very fast, towards hundreds of seconds

2. **What happens if Y is 1 or more at a junction?**
   - **A** ☐ The junction is over capacity: no cycle length can fix it
   - **B** ☐ DoS falls below 0.9
   - **C** ☐ Webster gives a perfect 60 s cycle
   - **D** ☐ Actuation solves it by ending greens on gaps

3. **Lost time L grows (longer intergreens), Y unchanged. What happens to Webster's C_0?**
   - **A** ☐ It becomes exactly L
   - **B** ☐ It stays the same
   - **C** ☐ It gets longer
   - **D** ☐ It gets shorter

4. **A fixed plan was designed from last year's counts, and the rush hour now brings more traffic. What happens?**
   - **A** ☐ The cycle shortens automatically
   - **B** ☐ The stored green cannot clear it, so a queue piles up
   - **C** ☐ The plan updates itself from the loops
   - **D** ☐ Delay falls

5. **In an actuated signal, the gap between cars at the stop-line loop gets long. What happens?**
   - **A** ☐ The offset is changed
   - **B** ☐ The green is extended to the maximum
   - **C** ☐ The cycle is lengthened for the whole region
   - **D** ☐ The green ends

6. **A platoon is 15 s away up the road. What can an actuated signal do about it?**
   - **A** ☐ Hold the green ready for it, using its model
   - **B** ☐ Shift its offset to meet it
   - **C** ☐ Lengthen the regional cycle
   - **D** ☐ Nothing yet: its stop-line loops cannot see what is coming

7. **In Lab F3, the traffic was light. What did SCOOT's cycle optimiser do?**
   - **A** ☐ Switched the junction to fixed time
   - **B** ☐ Swapped to a stored plan
   - **C** ☐ Lengthened it to the maximum
   - **D** ☐ Shortened the cycle

8. **y(A) rises while y(B) stays the same. In Webster's split $g_i = (C_0 - L) \times y_i \div Y$, what happens to A's share?**
   - **A** ☐ A gets a smaller share
   - **B** ☐ The shares stay equal
   - **C** ☐ A gets a bigger share of the green
   - **D** ☐ B gets all the green

9. **Junctions move from isolated to closely spaced along a busy corridor. What happens to SCOOT's advantage?**
   - **A** ☐ It stays exactly the same
   - **B** ☐ Fixed time always wins there
   - **C** ☐ It disappears
   - **D** ☐ It tends to grow, because coordination matters more

10. **SCOOT's upstream loops start giving wrong counts. What happens?**
    - **A** ☐ Delay falls because fewer cars are seen
    - **B** ☐ The model's predictions go wrong, and so do the timings
    - **C** ☐ Nothing: SCOOT ignores the loops
    - **D** ☐ SCOOT turns into actuated control by itself

GROUP 6 · SCOOT VS OTHERS · QUIZ 5 OF 6

Everyday pictures

Names: _____ Score: ____ / 10

1. **In the course analogy, a printed bus timetable stands for…**

   A ☐ Actuation
   B ☐ A pedestrian button
   C ☐ SCOOT
   D ☐ Fixed time

2. **“A driver who leaves as soon as nobody else is running for the bus” stands for…**

   A ☐ Fixed time
   B ☐ Webster’s formula
   C ☐ Actuation
   D ☐ SCOOT

3. **The dispatcher with a radio who hears a crowd is coming in 15 seconds stands for…**

   A ☐ SCOOT, warned by its upstream loops
   B ☐ Actuation, watching the stop line
   C ☐ The engineer with a stopwatch
   D ☐ Fixed time, reading its timetable

4. **Where does the dispatcher analogy break down?**

   A ☐ The dispatcher is always slower than the timetable
   B ☐ The timetable never works
   C ☐ The dispatcher cannot hear the radio
   D ☐ At a single junction the dispatcher’s advantage is small; it grows with many junctions

5. **Memory palace: a timetable robot finishes at 19, a jumpy rabbit at 14 and a half, a calm dispatcher at just under 13. What are those times?**

   A ☐ Corridor bus delays
   B ☐ Cycle times in seconds
   C ☐ Webster greens
   D ☐ Lab F3 mean delays: 18.96, 14.48, 12.79 s

6. **In the memory palace, the scales on the podium labelled “1.5 L + 5” remind you of…**

   A ☐ The LPU rule
   B ☐ The performance index
   C ☐ Robertson’s F
   D ☐ Webster’s optimum cycle formula

7. **The Lab F3 bar chart shows mean delay for the three controllers. Which bar is tallest?**

   A ☐ Actuated
   B ☐ SCOOT
   C ☐ All are equal
   D ☐ Fixed time

8. **The chart caption says the gap between SCOOT and actuation is small. What lesson does it draw?**

   A ☐ At an isolated junction actuation is strong; coordination is where SCOOT pays off
   B ☐ SCOOT failed the test
   C ☐ Fixed time is best in light traffic
   D ☐ Actuation is useless

9. **The “elastic plan” animation: a fixed 24 s green against SCOOT’s green in the morning rush. Like which everyday device does SCOOT behave?**

   A ☐ A thermostat, nudging a little at a time
   B ☐ A printed timetable
   C ☐ An alarm clock that rings once
   D ☐ A light switch, on or off

10. **In the bathtub picture, a fixed plan meets a rush hour with more traffic than its green can clear. What do you see?**

    A ☐ The tub fills more each cycle: the queue piles up
    B ☐ The plug is never put in
    C ☐ The tub empties faster than before
    D ☐ The tap stops

GROUP 6 · SCOOT VS OTHERS · QUIZ 6 OF 6

Exam challenge

Names: _____ Score: ____ / 10

1. **A fixed plan was built for Y = 0.60, L = 10 s (C_o = 50 s). Traffic grows to Y = 0.75. What would Webster now recommend?**

   A ☐ 200 s
   B ☐ 62.5 s
   C ☐ 50 s
   D ☐ 80 s

2. **Same junction: Y = 0.75 but the cycle stays 50 s with L = 10 s. Using DoS = Y × C ÷ (C − L), what is the DoS?**

   A ☐ 0.60
   B ☐ About 1.25
   C ☐ 0.75
   D ☐ About 0.94

3. **Which statement about the evidence is true?**

   A ☐ Actuation beat SCOOT in Lab F3
   B ☐ SCOOT is usually better, not always: in the corridor run actuation had slightly less mean delay
   C ☐ Fixed time won the corridor run
   D ☐ SCOOT had the lowest delay in every test

4. **A council's main goal is reliable buses on a busy corridor. Based on the corridor run, which controller suits it better and why?**

   A ☐ Actuated: 1.3 s less mean delay, so buses must be faster
   B ☐ Either: the results were identical
   C ☐ SCOOT: bus delay 22.9 s vs 38.0 s, with fewer stops
   D ☐ Fixed time: it never changes

5. **Why does SCOOT beat fixed time by much more (about 33%) than it beats actuation (about 12%) at one junction?**

   A ☐ SCOOT ran a longer cycle
   B ☐ Actuation already reacts to the traffic; fixed time cannot react at all
   C ☐ Fixed time had broken detectors
   D ☐ Actuation uses upstream loops too

6. **TRANSYT and SCOOT share the same profile and dispersion model. Which dispersion idea do they share (link to group 3)?**

   A ☐ The LPU rule
   B ☐ Webster's optimum cycle
   C ☐ Robertson's model, F = 1 ÷ (1 + αβT)
   D ☐ Plan selection

7. **Webster's plan is said to stay good "only while the flows stay the same". Which SCOOT feature answers this weakness?**

   A ☐ Removing all detectors
   B ☐ A fixed 120 s cycle
   C ☐ Small, frequent changes to split, cycle and offset from live upstream data
   D ☐ A bigger library of stored plans

8. **An engineer is told to make an isolated, lightly used junction work well cheaply. What is a fair choice?**

   A ☐ A simple actuated (or fixed) controller may be good enough
   B ☐ Set Y above 1
   C ☐ Only SCOOT can run it
   D ☐ Use Webster with Y = 0.95

9. **Webster with L = 14 s and Y = 0.60. What is C_o, and what is the green to share?**

   A ☐ 57 s and 43 s
   B ☐ 65 s and 65 s
   C ☐ 65 s and 51 s
   D ☐ 26 s and 12 s

10. **Why does an actuated signal not coordinate a green wave like SCOOT?**

A ☐ It decides locally from its own stop-line loops and does not share a cycle with neighbours

B ☐ It has no loops at all

C ☐ It copies SCOOT's offsets

D ☐ It always runs the longest cycle

Answer key

Group 1 · quiz 1 · Basics		
1	**D**	SCOOT = Split Cycle Offset Optimisation Technique: it adjusts those three timing variables.
2	**A**	The split shares the green; the cycle is the full round of stages; the offset links neighbouring junctions.
3	**D**	Offsets coordinate neighbours so a platoon released by one junction meets a green at the next (a green wave).
4	**D**	A few seconds before each stage change it tries changing it a little earlier, keeping it, or a little later.
5	**C**	The cycle optimiser works on a slower clock than the split and offset optimisers.
6	**A**	3600 ÷ 60 = 60 cycles; two stage changes per cycle; 60 × 2 = 120.
7	**A**	120 + 60 + 12 = 192 small timing decisions an hour at one junction.
8	**C**	Small, frequent changes are robust: errors stay small and the plan tracks demand like a thermostat.
9	**A**	A thermostat makes small corrections often. A printed timetable is the picture of a fixed-time plan.
10	**C**	Plan selection jumps between pre-made plans; SCOOT keeps one plan and stretches it gently, like elastic.

Group 1 · quiz 2 · Key words		
1	**A**	The cycle is one full round of stages, for example 60 s; the split shares it out and the offset links neighbours.
2	**A**	For example, the main road gets green together in one stage and the side road in the next.
3	**C**	The cycle is shared by all junctions in a region, so neighbours can keep their offsets and a green wave.
4	**B**	Intergreen is the changeover between stages; it is partly lost time, when nobody moves well.
5	**A**	Each link has its own detector and its own small traffic model; the junctions themselves are nodes.
6	**C**	One central computer runs a small model for every link every second and sends the new timings out to the signals.
7	**C**	The detector counts feed a small traffic model of each link; the model's profile, queue and DoS drive the optimisers.
8	**D**	SCOOT keeps editing the plan it is running, a few seconds at a time, like elastic that stretches but never snaps.
9	**B**	A good offset ≈ the travel time between junctions, so the platoon arrives just as the next light turns green.
10	**B**	Splits change at every stage change, offsets once per cycle, and the region's cycle only every 5 minutes (2.5 when rising).

Group 1 · quiz 3 · Numbers and calculations		
1	**C**	Offset decisions are once per cycle: 3600 ÷ 60 = 60. 120 is the split count for two stages; 12 is the cycle count.
2	**D**	3600 ÷ 80 = 45 cycles; 3 stage changes per cycle; 45 × 3 = 135. 45 forgets to multiply by the stages.
3	**B**	60 ÷ 2.5 = 24 cycle reviews an hour, twice the usual 60 ÷ 5 = 12.
4	**D**	3600 ÷ 90 = 40 cycles. Split 40 × 2 = 80, offset 40, cycle 60 ÷ 5 = 12: 80 + 40 + 12 = 132.
5	**C**	60 − 26 − 5 − 5 = 24 s. 29 s forgets one intergreen; 34 s forgets both.
6	**A**	3600 ÷ 100 = 36 cycles; 4 stage changes per cycle; 36 × 4 = 144.
7	**B**	Good offset ≈ travel time = 300 ÷ 15 = 20 s, so the second green starts about 20 s after the first.

8	**D**	Travel time = 400 ÷ 10 = 40 s, so the green at the second junction should start about 40 s after the first.
9	**D**	Each choice changes the stored plan by only 1 s: 22 − 3 × 1 = 19 s. 10 s wrongly applies the full 4 s each time (22 − 12).
10	**B**	12 ÷ 48 = 0.25 = 25%. A longer cycle with the same 12 s lost would waste a smaller share.

Group 1 · quiz 4 · What happens if…?		
1	**A**	The split optimiser runs before every stage change and moves green towards the stage that needs it, a few seconds at a time.
2	**A**	Cycles per hour fall from 3600 ÷ 60 = 60 to 3600 ÷ 90 = 40, so there are fewer stage changes and fewer offset checks.
3	**A**	SCOOT reviews the cycle every 5 minutes, and every 2.5 minutes while traffic is rising, so it keeps up with the peak.
4	**C**	The cycle is shared by the region; a longer cycle wastes a smaller share of time on stage changes and adds capacity.
5	**A**	The same 12 s is spread over a longer cycle: 12 ÷ 44 ≈ 27%, 12 ÷ 64 ≈ 19%, 12 ÷ 88 ≈ 14%.
6	**D**	A fixed plan cannot react; when arrivals exceed what the green can clear (DoS above 1), the queue grows cycle after cycle.
7	**A**	When the peak eases the cycle optimiser holds or shortens the cycle, step by step.
8	**C**	In the course animation the offset walks 8 s → 16 s → 20 s and stops at J2 fall from 5 to 0, with the PI falling from 230 to 94.
9	**A**	Each nudge is small and there are about 192 decisions an hour at one junction, so one wrong step is quickly put right.
10	**B**	The trial change (−4/0/+4 s) is used now, but the permanent change is only 1 s, for example 30 → 31 s.

Group 1 · quiz 5 · Everyday pictures		
1	**A**	The analogy fits the small, frequent nudges, but SCOOT has split, cycle and offset to turn, each on its own clock.
2	**B**	Jumping between whole stored settings is like plan selection; the thermostat's many small nudges are the elastic plan.
3	**C**	The band stretches but never snaps, like the timings that change a little at a time; the dials S, C and O are the three knobs.
4	**C**	Many small twangs, never a snap: small, frequent changes that follow the traffic.
5	**B**	The teacher gives each row (stage) its turn, over and over; one full round of rows is the cycle.
6	**B**	Finish times are the Lab F3 delays: fixed 18.96 s (robot), actuated 14.48 s (rabbit), SCOOT 12.79 s (dispatcher).
7	**C**	The fixed plan keeps the same green; SCOOT's green grows and shrinks with the traffic instead.
8	**A**	26 + 5 + 24 + 5 = 60 s: one full round of stages is the cycle; how the 50 s of green is shared is the split.
9	**A**	Detector → LPUs each second → link model → split, offset and cycle optimisers → new timings → drivers → detector again.
10	**D**	Each line is a car; as the offset walks to about 20 s, the lines reach J2 on green and the stops there fall to 0.

Group 1 · quiz 6 · Exam challenge		
1	**A**	Split 60 × 2 = 120, offset 60, cycle 60 ÷ 2.5 = 24: 120 + 60 + 24 = 204. 192 uses the normal 5-minute review.

2	**C**	Fastest to slowest: split before every stage change, offset once per cycle, the region's cycle every 5 minutes (2.5 when rising).
3	**A**	An offset only means something if both junctions repeat at the same rhythm; different cycles would slide past each other.
4	**C**	Capacity = 1800 × 25 ÷ 60 = 750 veh/h; DoS = 540 ÷ 750 = 0.72, comfortably below 0.9. 1.39 is the inverted ratio.
5	**A**	0.78 × 64 ÷ (64 − 12) = 49.92 ÷ 52 = 0.96, down from 1.07 at 44 s. 0.81 is the later 88 s value with Y = 0.70.
6	**D**	The split optimiser picks the option with the lowest worst DoS (0.71), and the permanent change is only 1 s: 30 → 31 s.
7	**D**	(18.96 − 12.79) ÷ 18.96 = 6.17 ÷ 18.96 ≈ 0.33. 48% divides by 12.79 instead; 12% is SCOOT against actuated.
8	**B**	Actuation reacts only locally, ending the green on a gap; SCOOT predicts from upstream and coordinates neighbours.
9	**D**	A car crossing the upstream loop reaches the stop line 10–20 s later, time enough for the optimisers to act.
10	**B**	Mean delay was slightly higher with SCOOT, while stops and bus delay fell; judge a system on several measures, not one.

Group 2 · quiz 1 · Basics		
1	**A**	Upstream placement means cars are counted long before they reach the stop line.
2	**A**	The warning lets the model predict arrivals and get the green ready in time.
3	**B**	Each quarter second the loop reports whether it is covered.
4	**C**	LPUs are SCOOT's own unit for an amount of traffic, mixing a count with occupancy.
5	**A**	One unit for being a vehicle, plus two for every second it blocks the loop. (The real TRL coding differs; this is the lab's teaching rule.)
6	**C**	Covered time = (vehicle length + loop length) ÷ speed = 6.5 ÷ 13.5 = 0.48 s.
7	**D**	1 + 2 × 0.48 = 1.96 LPU.
8	**D**	Covered (12 + 2) ÷ 13.5 = 1.04 s, so 1 + 2 × 1.04 = 3.07 LPU.
9	**D**	Slow traffic covers the loop longer, so LPUs rise: an early sign of congestion.
10	**D**	A bus needs more green than a car; counting both as "1" would hide that.

Group 2 · quiz 2 · Key words		
1	**A**	The loop is a buried coil of wire; metal passing over it switches it on, so SCOOT knows when it is covered.
2	**A**	Occupancy is the covered time (or the share of time the detector is covered); long or slow vehicles cover it longer.
3	**B**	SCOOT sees a town as junctions joined by one-way approach roads; each approach is a link with its own detector and model.
4	**B**	Nodes are the signalled junctions; links are the approach roads between them.
5	**C**	The upstream green lets a bunch of cars go together; that bunch is the platoon SCOOT counts at the loop.
6	**C**	T is the detector-to-stop-line travel time, for example 140 m at 14 m/s = 10 s.
7	**C**	Upstream is the end of the link traffic enters from, so the loop sees cars long before they reach the lights.
8	**D**	Flow counts how many pass, in veh/h or veh/s; headway and occupancy are the other two descriptions.
9	**B**	Headway = 1 ÷ flow; for 0.15 veh/s it is about 6.7 s.

10	**C**	Actuated loops at the stop line extend the green while cars keep coming; SCOOT puts its loops upstream, and fixed time needs none.

Group 2 · quiz 3 · Numbers and calculations		
1	**C**	Covered (4 + 2) ÷ 12 = 0.5 s; LPU = 1 + 2 × 0.5 = **2.00**. 1.50 forgets the ×2; 1.67 forgets the loop length; 1.00 forgets occupancy.
2	**D**	Occupancy = (8 + 2) ÷ 5 = 10 ÷ 5 = **2.0 s**. 1.6 s forgets the loop; 0.5 s divides the wrong way; 50 s multiplies.
3	**C**	Covered (16.5 + 2) ÷ 9.25 = 18.5 ÷ 9.25 = 2.0 s; LPU = 1 + 2 × 2.0 = **5.0**. 3.0 forgets the ×2; 4.57 forgets the loop length; 2.0 is only the covered time.
4	**D**	3.07 ÷ 1.96 = **1.57**. 0.64 is the ratio upside down; 1.11 is the difference; 2.67 is just the length ratio 12 ÷ 4.5.
5	**B**	10 × 1.95 = **19.5 LPU**, the queue used in the validation animation. 10 confuses vehicles with LPUs.
6	**C**	Warning = cruise time = 180 ÷ 12 = **15 s**, inside the usual 10–20 s. 0.07 s divides the wrong way; 2160 s multiplies.
7	**C**	1.5 s × 4 readings per second = **6** readings. 0.375 divides instead of multiplying.
8	**A**	LPU = 1 × 2 vehicles + 2 × 0.5 s = 2 + 1 = **3.0**. 2.0 counts only vehicles; 2.5 forgets the ×2.
9	**B**	36 km/h = 36 ÷ 3.6 = 10 m/s; covered 6.5 ÷ 10 = 0.65 s; LPU = 1 + 2 × 0.65 = **2.30**. 1.36 forgets to convert km/h; 1.90 forgets the loop; 1.65 forgets the ×2.
10	**C**	Covered 14 ÷ 6.75 = 2.07 s; LPU = 1 + 2 × 2.07 = **5.15**. 3.07 is the old speed; 6.14 doubles the whole LPU (the 1 for the vehicle does not double); 2.07 is only the covered time.

Group 2 · quiz 4 · What happens if…?		
1	**B**	Occupancy = (length + loop) ÷ speed, so double speed halves it; the "1 per vehicle" part stays, so LPU falls towards 1.
2	**B**	Loop length is in the occupancy formula: (vehicle + loop) ÷ speed, so a longer loop means more seconds covered and more LPUs.
3	**B**	Upstream loops see cars 10–20 s before they reach the lights; a stop-line loop sits under the queue and sees arrivals too late to predict.
4	**B**	Occupancy is what makes long vehicles score more; without it a bus counts as "1", like a car, though it needs more green.
5	**A**	Longer vehicle, longer covered time: (7 + 2) ÷ 13.5 = 0.67 s, so 1 + 2 × 0.67 = 2.33. 3.74 is the 16.5 m artic lorry.
6	**C**	Warning = cruise time from loop to stop line; a short distance leaves little time to predict and get the green ready.
7	**B**	Each lorry covers the loop longer and scores more LPUs, so demand in LPU rises even though the count does not, which matches the extra green lorries use.
8	**A**	Lower speed means distance ÷ speed is larger, so T grows; occupancy = (length + loop) ÷ speed also grows, so LPUs rise.
9	**A**	A car is about 2 LPU in this lab, so 10 cars ≈ 20 LPU. Always check vehicles versus LPUs before comparing.
10	**C**	The vehicle still counts 1; it is the covered time (12 + 2) ÷ speed that grows when the bus is slower, so its LPU rises.

Group 2 · quiz 5 · Everyday pictures		
1	**C**	LPU = 1 × vehicle + 2 × seconds covered; the entry coin is the count part.
2	**D**	The blocking charge is the occupancy part of the LPU: two units per second covered.

3	**A**	The bill measures how much of the doorway was used, not just how many came in; a bus scores about 3 LPU, a car about 2.
4	**D**	A slow car covers the loop longer and scores more LPUs, an early sign of congestion.
5	**C**	The mat is far from the lights, like SCOOT's upstream loop, which gives 10–20 s of warning.
6	**B**	SCOOT never sees the queue directly; the upstream loop is its early-warning radio.
7	**C**	The loop sits just after the upstream junction, so the model can predict when the cars arrive and get the green ready.
8	**B**	Car covered 0.48 s, bus 1.04 s; LPU = 1 + 2 × covered time gives 1.96 and 3.07.
9	**C**	Longer vehicles cover the 2 m loop longer, so each bar is taller than the one before.
10	**A**	Occupancy grows with vehicle length; about 1.05 s matches the 12 m bus at 13.5 m/s (1.04 s), while 0.5 s matches a car.

Group 2 · quiz 6 · Exam challenge		
1	**B**	An average so close to one car's 1.96 means most vehicles behaved like the 4.5 m car at about 13.5 m/s.
2	**A**	Actuated reacts locally at the stop line; SCOOT counts upstream, predicts, and makes small changes across a region.
3	**D**	Occupancy = (vehicle length + loop length) ÷ speed: longer vehicles and lower speeds both mean more seconds covered.
4	**C**	12 × 1.96 + 2 × 3.07 = 23.52 + 6.14 = **29.66 LPU**. 14 counts vehicles; 28 treats every vehicle as 2 LPU; 23.52 leaves out the buses.
5	**A**	Lag = βT = 0.8 × 10 = **8 s**: the platoon's fastest cars arrive before the average cruise time T.
6	**B**	Arrivals 9 × 2 = 18 LPU; capacity 25 × 0.8 = 20 LPU; DoS = 18 ÷ 20 = **0.90**, the SCOOT target. 0.45 uses 9 vehicles instead of 18 LPU; 1.11 is upside down.
7	**C**	SATO is in LPU/s, so the queue must be in LPU too: 10 cars ≈ 19.5 LPU, which takes much longer to clear.
8	**C**	3.0 = 1 + 2 × covered, so covered = 1.0 s; speed = (4.5 + 2) ÷ 1.0 = **6.5 m/s**. 3.25 m/s would give 2.0 s covered; 13.5 m/s gives only 1.96 LPU.
9	**B**	540 ÷ 3600 = 0.15 veh/s; 0.15 × 1.96 ≈ **0.29 LPU/s**. 0.15 forgets to convert to LPU; 6.7 is the headway in seconds.
10	**C**	Upstream detection gives 10–20 s of warning so SCOOT predicts arrivals; actuated control only reacts at the stop line.

Group 3 · quiz 1 · Basics		
1	**D**	When the upstream light turns green, cars leave together as a platoon.
2	**B**	One bar per second of the cycle; the platoon shows up as a block of tall bars.
3	**A**	Fast drivers pull ahead and slow drivers fall behind, so the bunch gets longer and flatter.
4	**C**	It is measured on the street during validation.
5	**C**	Lag = β × T = 0.8 × 10 = 8 s: the fastest drivers arrive before the average car.
6	**D**	1 ÷ (1 + 0.35 × 0.8 × 10) = 1 ÷ 3.8 = 0.263.
7	**D**	Each second: arrivals = 0.263 × new traffic + 0.737 × last second's arrivals. Nobody is lost.
8	**C**	1 ÷ (1 + 0.35 × 0.8 × 20) = 1 ÷ 6.6 ≈ 0.15. The longer the trip, the more the bunch smears out.
9	**B**	0.263 × 1 + 0.737 × 0.263 = 0.263 + 0.194 = 0.457.
10	**C**	The total is unchanged (10 LPU in the example); it is only rearranged in time, over about 25 s instead of 10 s.

Group 3 · quiz 2 · Key words		
1	**B**	Fast drivers pull ahead and slow ones fall behind, so the bunch arrives longer and flatter; no cars are lost.
2	**C**	SCOOT shifts and smooths the profile with Robertson's model, the same one used in the TRANSYT design program.
3	**D**	TRANSYT designs fixed plans for networks off-line; SCOOT is often called an 'on-line TRANSYT'.
4	**C**	Each second: arrivals = F × detected + (1 − F) × last second's arrivals. A small F means a lot of spreading.
5	**B**	The glossary gives α = 0.35 and β = 0.8; the model uses fixed constants that can be tuned to the road.
6	**A**	The front of the platoon arrives at βT = 0.8 T, earlier than the average car, because the fastest drivers lead.
7	**A**	SCOOT sees a town as nodes (junctions) joined by one-way links, each with its own detector and model.
8	**D**	One bar for each second of the cycle; a platoon released by the same upstream green appears in the same place every cycle.
9	**C**	One odd cycle can move a bar only a quarter of the way; the regular platoon stands out while noise fades.
10	**A**	The trap note says the front arrives at 0.8 T and the rest arrive later in the tail; the curve fades away over about 15 s.

Group 3 · quiz 3 · Numbers and calculations		
1	**D**	T = 180 ÷ 12 = 15 s; lag = 0.8 × 15 = 12 s. The lag is not the full T; dividing 12 ÷ 180 inverts the ratio.
2	**B**	F = 1 ÷ (1 + 0.35 × 0.8 × 5) = 1 ÷ 2.4 = 0.417. 0.364 forgets β, 0.583 is 1 − F, 0.200 is 1 ÷ T.
3	**D**	F = 1 ÷ (1 + 0.35 × 0.8 × 25) = 1 ÷ (1 + 7) = 0.125. 0.103 forgets β, 0.875 is 1 − F, 0.143 = 1 ÷ 7 forgets the 1 +.
4	**C**	Lag = β × T = 0.8 × 20 = 16 s. 20 s is the full cruise time, 25 s divides by 0.8, 4 s is 0.2 × T.
5	**A**	0.263 × 1 + 0.737 × 0.457 = 0.263 + 0.337 = 0.600. 0.337 forgets the new traffic; 0.720 forgets to multiply 0.457 by 0.737.
6	**A**	0.263 × 0 + 0.737 × 0.95 ≈ 0.70. The arrivals fade gradually; they do not drop to zero at once. 0.25 uses F instead of 1 − F.
7	**C**	0.75 × 4 + 0.25 × 8 = 3 + 2 = 5.0. 6.0 is a plain average; 7.0 swaps the weights; 8.0 ignores the history.
8	**B**	0.5 × 2 + (1 − 0.5) × 1 = 1 + 0.5 = 1.5. 3.0 adds without the shares F and 1 − F.
9	**C**	Total = 2 × 6 = 12 LPU. Robertson's recipe only rearranges arrivals in time; the area under the curve is unchanged.
10	**B**	0.263 × 2 + 0.737 × 0 = 0.526. 0.263 forgets the 2 LPU; 1.474 uses 1 − F instead of F.

Group 3 · quiz 4 · What happens if…?		
1	**B**	F = 1 ÷ (1 + 0.28 T): a smaller T gives a bigger F, so less of each second is smeared into later seconds.
2	**C**	The lag is 0.8 × T, so it grows with T; for example T = 10 → 8 s, T = 20 → 16 s.
3	**C**	F = 1 ÷ (1 + αβT): a bigger α makes the bottom bigger, so F is smaller and the spreading larger. The total stays the same.
4	**C**	With no new traffic, arrivals = (1 − F) × previous = 0.737 × previous, so the tail decays over about 15 s.
5	**D**	Each step is F × detected + (1 − F) × previous, so doubling every detected value doubles every arrival (second 1 becomes 0.526 instead of 0.263).
6	**C**	New bar = 0.75 × old + 0.25 × this cycle, so one odd cycle shifts it only 25% of the way.
7	**A**	Cars need about the cruise time to reach the stop line; the model shifts the profile forward (front at 0.8 T) to match when they really arrive.

8	**A**	The green wave works best when platoons stay tight; on a long road the platoon has smeared out, so the offset matters less.
9	**B**	F = 1 ÷ (1 + 0.28 × 0) = 1, so arrivals = 1 × detected + 0 × previous: no shift and no smear.
10	**C**	Nothing is lost, only rearranged in time: the 10 LPU of the worked example still arrive, over about 25 s. 0.95 is the peak height, not the total.

Group 3 · quiz 5 · Everyday pictures		
1	**A**	The class leaves together but arrives over a longer time, just like a platoon between detector and stop line.
2	**D**	The class leaves the gate (the upstream junction and loop) and is spread out by the time it reaches the bus stop (the stop line).
3	**C**	In the worked example the peak falls from 1.0 to 0.95 and the platoon lasts about 25 s instead of 10 s.
4	**A**	Real dispersion depends on the road, so engineers can tune the fixed constants α and β.
5	**C**	35 players → α = 0.35; 80% speed → β = 0.8. The band spreading out is dispersion.
6	**C**	The band leaves in a tight block and arrives at the far end spread out, with trombones early and drummers trailing.
7	**B**	The caption says: same cars, longer time, lower peak; a longer road means more spreading.
8	**C**	Each second's arrival is a share F of the new traffic plus the rest of last second's arrival, like a running average.
9	**A**	The front arrives at 0.8 × 10 = 8 s; the area under both curves is the same 10 LPU.
10	**D**	Each new cycle is blended in at 25%, so the platoon's place in the cycle stands out while random noise fades.

Group 3 · quiz 6 · Exam challenge		
1	**B**	0.263 + 0.457 + 0.600 = 1.32 LPU, while 3 LPU have crossed the loop: the rest is still to come in later seconds. 0.79 = 3 × 0.263 forgets the carry-over.
2	**C**	T = 300 ÷ 15 = 20 s; lag = 0.8 × 20 = 16 s; F = 1 ÷ (1 + 0.35 × 0.8 × 20) = 1 ÷ 6.6 = 0.152. 0.848 is 1 − F.
3	**A**	The lag is 0.8 T, not T (a listed confusion). The SCOOT loop is upstream, not at the stop line.
4	**D**	The smaller F, the less of each second arrives at once and the more spreading; F falls as T grows.
5	**B**	With steady 1 LPU/s the arrivals would creep towards 1.0 ($1 - 0.737^{10} \approx 0.95$ after 10 s), but the input stops first. Nothing is lost.
6	**D**	Cycle 1: 0.75 × 4 + 0.25 × 8 = 5.0. Cycle 2: 0.75 × 5 + 0.25 × 8 = 3.75 + 2 = 5.75. A lasting change is followed, but gently.
7	**D**	The total is conserved, so arrivals per cycle in the DoS are unchanged; the timing within the cycle changes, which is what offsets and stops depend on.
8	**A**	Good offset ≈ travel time; the model shifts the profile (front at 8 s) and spreads it so the offset optimiser judges delay and stops from true arrivals.
9	**B**	Lag = 0.8 T gives 8 s and 16 s; F = 1 ÷ (1 + 0.28 T) gives 0.263 and 0.152. Longer link → later and more spread.
10	**A**	The queue model fills the bathtub from the predicted arrivals and drains it at SATO during green; SCOOT models the queue from the upstream counts.

Group 4 · quiz 1 · Basics		
1	**B**	Arrivals pour in all the time; red is the plug; the plughole is the green discharge (SATO).
2	**C**	It is measured in LPU per second of green, and checked on site during validation.

3	**D**	0.2 × 35 = 7 LPU.
4	**D**	Net drain = 0.6 − 0.2 = 0.4 LPU/s.
5	**B**	7 ÷ 0.4 = 17.5 s after the green starts.
6	**B**	DoS = (arrivals per cycle + leftover queue) ÷ (effective green × SATO).
7	**B**	12 ÷ 15 = 0.80 (80%): below the 90% target, so there is spare green.
8	**B**	Traffic varies from cycle to cycle; a 10% margin stops a single busy cycle from building a queue.
9	**A**	Above 1.0 more arrives than the green can clear, so the leftover queue grows each cycle.
10	**C**	Every vehicle-second spent waiting adds to the area; in the example it is 183.75 LPU·s per cycle.

Group 4 · quiz 2 · Key words		
1	**C**	Effective green = green − start lag + end lag. DoS uses this, not the displayed green.
2	**A**	Drivers need a moment to react and get going, so effective green begins a start lag late (3 s in the course's SATO animation).
3	**C**	Drivers keep crossing during amber, so effective green runs an end lag past the end of the displayed green.
4	**A**	It is added to the next cycle's demand in the DoS: (arrivals per cycle + residual) ÷ (effective green × SATO).
5	**D**	Anyone arriving before the queue has cleared has to halt; in the worked example that is 10.5 LPU per cycle.
6	**D**	PI = delay + K × stops; lower is better, and the optimisers compare options using it.
7	**B**	A bigger K makes stops count for more in the PI, so options with fewer stops look better.
8	**B**	Delay is the area under the queue curve: queue (LPU) × time (s), for example 183.75 LPU·s per cycle.
9	**A**	About one car every 2 s per lane on a good lane; SCOOT measures the same idea in LPU/s and calls it SATO.
10	**C**	Capacity = saturation flow × green ÷ cycle; for example 1800 × 25 ÷ 60 = 750 veh/h.

Group 4 · quiz 3 · Numbers and calculations		
1	**C**	Queue = arrival rate × red = 0.25 × 40 = **10 LPU**. 160 comes from dividing 40 ÷ 0.25 instead of multiplying.
2	**C**	Net drain = 0.75 − 0.25 = 0.5 LPU/s, so 10 ÷ 0.5 = **20 s**. 13.3 s forgets that cars keep arriving (10 ÷ 0.75).
3	**A**	Demand = 0.15 × 80 = 12 LPU; capacity = 40 × 0.5 = 20 LPU; DoS = 12 ÷ 20 = **0.60**. 1.67 is the ratio upside down.
4	**B**	DoS = (15 + 3) ÷ (25 × 0.8) = 18 ÷ 20 = **0.90**. 0.75 forgets the residual queue (15 ÷ 20).
5	**A**	Effective green = green − start lag + end lag = 30 − 3 + 2 = **29 s**. 25 s subtracts both lags.
6	**C**	Two triangles: ½ × 30 × 6 + ½ × 20 × 6 = 90 + 60 = **150 LPU·s**. 90 counts only the red part; 300 forgets the ½.
7	**C**	PI = delay + K × stops = 150 + 10 × 10 = **250**. 160 forgets the weight K.
8	**A**	Stops = everyone who arrives before the queue clears = 0.2 × (35 + 17.5) = **10.5 LPU**. 7 LPU counts only red arrivals.
9	**D**	Capacity = 1800 × 25 ÷ 60 = 750 veh/h; DoS = 540 ÷ 750 = **0.72**. 1.39 is upside down; 0.30 forgets the green share.
10	**C**	New capacity = 15 + 4 × 0.6 = 17.4 LPU; DoS = 14.4 ÷ 17.4 ≈ **0.83**. 0.76 adds 4 LPU instead of 4 s × SATO.

Group 4 · quiz 4 · What happens if…?		
1	**C**	More demand over the same capacity: in the course example 0.2 → 0.24 LPU/s moves DoS from 0.80 to 0.96.
2	**C**	Capacity = effective green × SATO grows, so demand ÷ capacity falls (0.96 → 0.83 in the course animation).

3	**A**	SATO is the drain rate; a bigger drain clears the same queue in less time.
4	**D**	DoS = (arrivals per cycle + residual) ÷ (effective green × SATO), so a residual raises the top line.
5	**C**	Effective green = green − start lag + end lag, so a longer start lag cuts capacity and raises DoS.
6	**B**	The queue shrinks at SATO − arrival rate; if that is negative the tub fills even with the plug out.
7	**B**	Both the peak queue and the clearing time grow, so the area under the queue line grows.
8	**C**	PI = delay + K × stops; a bigger K makes each stop cost more in the score.
9	**C**	A bigger modelled drain empties the model tub too soon, so the engineer would turn SATO down during validation.
10	**B**	Spare green on a quiet link can go to the busiest stage at the junction, in small steps.

Group 4 · quiz 5 · Everyday pictures		
1	**C**	During red the plug is in, so the water (the queue) rises; on green the plug pops out.
2	**C**	Water rises in red and drains in green, just like the modelled queue.
3	**A**	If arrivals are slower than SATO, the queue shrinks at the difference and clears during green.
4	**A**	The basic bathtub ignores the queue's physical length; that comes back at the Advanced level (congestion).
5	**A**	The 90% mark is the DoS target; the water's area over time is the delay.
6	**A**	A thermostat reacts to the temperature reading; SCOOT reacts to how full the green is and adjusts a few seconds at a time.
7	**C**	Fill rate = arrivals 0.2 LPU/s; drain rate = SATO − arrivals = 0.6 − 0.2 = 0.4 LPU/s.
8	**B**	It clears 17.5 s into a 25 s green, so 25 − 17.5 = 7.5 s are empty: spare green.
9	**B**	More green raises capacity: 14.4 ÷ 15.0 = 0.96 becomes 14.4 ÷ 17.4 ≈ 0.83.
10	**D**	A bigger plughole drains faster; the slope of the cleared-LPU line is SATO.

Group 4 · quiz 6 · Exam challenge		
1	**A**	Effective green = 27 − 3 + 1 = 25 s; capacity 25 × 0.6 = 15 LPU; DoS = 12 ÷ 15 = **0.80**. 0.74 wrongly uses the displayed 27 s.
2	**A**	Need 12 ÷ (g × 0.6) = 0.9, so g = 12 ÷ (0.9 × 0.6) ≈ **22.2 s**. 20 s would give exactly 100%.
3	**C**	Each cycle leaves 18 − 15 = 3 LPU more, so 3 × 3 = **9 LPU**: above 100% the queue grows every cycle.
4	**D**	100% is not gridlock; it means no margin. That is why SCOOT aims for about 90%.
5	**D**	PI A = 200 + 10 × 12 = 320; PI B = 180 + 10 × 15 = 330. Lower PI wins, so **A**. 195 and 212 forget K.
6	**D**	SATO = queue ÷ (clear time − start lag) = 18 ÷ (33 − 3) = **0.60 LPU/s**. 0.55 forgets the start lag.
7	**B**	Capacity = 1800 × 30 ÷ 90 = 600 veh/h; DoS = 480 ÷ 600 = **0.80**.
8	**B**	Queue 0.2 × 70 = 14 LPU, clears in 14 ÷ 0.4 = 35 s; delay = ½ × 70 × 14 + ½ × 35 × 14 = 490 + 245 = **735** = 4 × 183.75. Both triangle sides double.
9	**D**	A longer cycle wastes a smaller share on stage changes, adding capacity; the cycle optimiser runs every 5 min, 2.5 when rising.
10	**B**	DoS = (12 + 1.5) ÷ 15 = 13.5 ÷ 15 = **0.90**. 0.80 forgets the residual queue.

Group 5 · quiz 1 · Basics		
1	**A**	The model is only as good as its settings, and every street is different.
2	**B**	The street is always right: the model is changed to match it.

3	**D**	Take away the start lag, then divide the queue by the time it took to drain.
4	**D**	19.5 ÷ (36 − 3) = 19.5 ÷ 33 = 0.59 LPU/s.
5	**B**	3 + 19.5 ÷ 0.67 = 3 + 29.1 = 32.1 s: 4 s too early compared with the street.
6	**C**	An optimistic plughole means the model under-estimates how much green the link needs.
7	**A**	The first cars react and accelerate before the queue flows at full rate.
8	**B**	Engineers also check the maximum queue clear time on long links.
9	**D**	Anything that slows the queue leaving the stop line lowers the drain rate.
10	**C**	A well-built guitar still plays wrong notes until it is tuned; a well-built model needs tuning to its street.

Group 5 · quiz 2 · Key words		
1	**D**	SATO is the queue discharge rate in LPU/s: the size of the plughole in the bathtub picture.
2	**C**	SATO is measured in LPU/s; the lab values are 0.67 LPU/s (main road) and 0.89 LPU/s (side road) per lane.
3	**D**	Cruise time T is the detector-to-stop-line journey time, for example 140 m at 14 m/s gives 10 s.
4	**B**	Effective green starts a start lag late and runs an end lag into amber; in Lab F2 the end lag is 4 s.
5	**D**	The queue only flows at SATO once drivers react, and some keep going into the amber.
6	**C**	The five street-dependent settings are cruise time, SATO, start lag, end lag and maximum queue clear time.
7	**A**	Queue clear time = start lag + queue ÷ SATO, for a queue standing at the start of green.
8	**A**	SCOOT never sees the real queue; it models it, which is why the model must be checked against the street.
9	**B**	When a link is commissioned, an engineer stands at the stop line and compares what they see with what SCOOT shows.
10	**C**	Saturation flow is about 1800 veh/h per lane; SCOOT measures the same top rate in LPU/s and calls it SATO.

Group 5 · quiz 3 · Numbers and calculations		
1	**B**	SATO = 18 ÷ (27 − 3) = 18 ÷ 24 = 0.75 LPU/s. Forgetting the start lag gives 18 ÷ 27 = 0.67; 1.33 is the ratio upside down.
2	**A**	SATO = 18 ÷ (30 − 3) = 18 ÷ 27 ≈ 0.67 LPU/s: too low, so the model cleared later than the street.
3	**A**	10 × 1.95 = 19.5 LPU: the queue used in the course's stopwatch example.
4	**A**	(0.67 − 0.59) ÷ 0.67 = 0.08 ÷ 0.67 ≈ 0.12 = 12%. 8% would use 0.08 as if it were a percentage.
5	**B**	3 + 19.5 ÷ 0.89 = 3 + 21.9 ≈ 24.9 s. 22 s forgets the start lag; 32 s uses the main-road SATO 0.67.
6	**A**	Clear time = 2 + 12 ÷ 0.6 = 2 + 20 = 22 s. 20 s forgets the start lag; 9.2 s multiplies instead of dividing.
7	**C**	SATO = 20 ÷ (28 − 3) = 20 ÷ 25 = 0.8 LPU/s. 0.71 forgets the start lag (20 ÷ 28); 1.25 is upside down.
8	**D**	Cruise time = distance ÷ speed = 162 ÷ 13.5 = 12 s. Multiplying gives the nonsense 2187.
9	**D**	Effective green = 30 − 3 + 4 = 31 s. 23 s subtracts both lags; 37 s adds both.
10	**A**	Model: 3 + 15 ÷ 0.6 = 28 s, 5 s too early. Corrected SATO = 15 ÷ (33 − 3) = 15 ÷ 30 = 0.5 LPU/s. 0.45 forgets the start lag.

Group 5 · quiz 4 · What happens if…?		
1	**B**	Clearing early means the model's plughole is too big, so SATO is too high and must be reduced.
2	**B**	If the model empties later than the street, it thinks traffic leaves more slowly than it really does.

3	**D**	Lower SATO makes the link look busier (higher DoS), so green that other movements could use is spent here.
4	**A**	Narrow lanes and parking slow the discharge; unlike a guitar, a link needs re-tuning when the layout changes.
5	**A**	Clear time = start lag + queue ÷ SATO, so adding 2 s to the start lag adds 2 s to the clear time.
6	**C**	For 19.5 LPU at 0.67: 3 + 29.1 = 32.1 s; for 39 LPU: 3 + 58.2 = 61.2 s. Only the drain part doubles.
7	**C**	DoS = (arrivals + residual) ÷ (effective green × SATO); a bigger SATO makes the denominator bigger.
8	**C**	Effective green runs an end lag into amber; a longer end lag means more real green than the model thinks.
9	**A**	Cruise time tells the model when detected cars reach the stop line, so it must match the street.
10	**C**	32.1 s instead of 36 s: small parameter errors give big errors in delay and DoS.

Group 5 · quiz 5 · Everyday pictures		
1	**C**	The model is the guitar; the street is the tuning fork you tune it to.
2	**C**	You pluck, listen and turn the peg; the engineer watches, compares and adjusts SATO and the lags.
3	**B**	Street changes alter SATO and lags, so validation is not a once-and-forever job.
4	**A**	Poor validation is the most common reason SCOOT underperforms.
5	**A**	The Lab F2 lags were a start lag of 3 s and an end lag of 4 s.
6	**C**	The stopwatch measures when the real queue clears, which is compared with the model.
7	**B**	Arrivals are the tap; in green the plughole drains the queue at SATO.
8	**D**	The caption says the slope of the cleared-LPU line is SATO; the flat start is the 3 s start lag.
9	**B**	A bigger SATO drains the same queue faster (a car every 2 s instead of every 3.3 s).
10	**D**	The model never sees the real queue, yet it follows it in timing and size; the small gap is about how SUMO counts.

Group 5 · quiz 6 · Exam challenge		
1	**B**	18 ÷ (30 × 0.67) = 18 ÷ 20.1 ≈ 0.90; 18 ÷ (30 × 0.59) = 18 ÷ 17.7 ≈ 1.02. The wrong SATO hid an overload.
2	**C**	Do not ‘fix’ a model by changing the target DoS or the optimiser. Validate the model first.
3	**B**	Queue = 10 × 1.95 = 19.5 LPU; clear time = 3 + 19.5 ÷ 0.65 = 3 + 30 = 33 s. 30 s forgets the start lag.
4	**D**	19.5 ÷ 0.67 ≈ 29.1 s; 19.5 ÷ 0.89 ≈ 21.9 s; difference ≈ 7.2 s. The start lag is the same on both, so it cancels.
5	**D**	T = 135 ÷ 13.5 = 10 s; F = 1 ÷ (1 + 0.35 × 0.8 × 10) = 1 ÷ 3.8 ≈ 0.263. A wrong cruise time would also give a wrong F.
6	**B**	Effective green = 25 − 3 + 4 = 26 s; 26 × 0.59 ≈ 15.3 LPU. Using 25 s gives 14.8; dividing gives 44.
7	**C**	That is why Lab F2 found 0.67 on the main road but 0.89 on the side road.
8	**B**	If the model empties first, it thinks traffic leaves faster than it does, so SATO must come down.
9	**A**	SCOOT trusts its model queue; an optimistic drain rate under-estimates the green needed.
10	**A**	SATO = 24 ÷ (33 − 3) = 24 ÷ 30 = 0.8 LPU/s, above 0.67, so the model was clearing too late. 0.73 forgets the start lag.

Group 6 · quiz 1 · Basics		
1	**A**	Fixed-time plans are often prepared off-line, for example with Webster’s formula or TRANSYT.
2	**B**	Actuation reacts to the traffic in front of it, but only locally.

3	**A**	SCOOT predicts from upstream and adapts split, cycle and offset across a region.
4	**B**	Fixed 18.96 s, actuated 14.48 s, SCOOT 12.79 s per vehicle.
5	**A**	(18.96 − 12.79) ÷ 18.96 = 6.17 ÷ 18.96 ≈ 0.33.
6	**D**	(14.48 − 12.79) ÷ 14.48 = 1.69 ÷ 14.48 ≈ 0.12.
7	**A**	SCOOT is usually better, not always: here actuation edged it on mean delay.
8	**D**	Stops 1.10 vs 1.15 per vehicle, and bus delay 22.9 s vs 38.0 s.
9	**D**	Adaptive control pays off most where traffic changes and junctions interact.
10	**D**	With faulty loops or untuned settings the model's predictions — and the timings — go wrong.

Group 6 · quiz 2 · Key words		
1	**A**	TRANSYT designs fixed plans for networks off-line, using the same profile and dispersion model as SCOOT.
2	**B**	TRANSYT does the modelling once, off-line; SCOOT does the same kind of modelling continuously, on-line.
3	**B**	L is the lost time per cycle (seconds when nobody moves well, mostly in the intergreens).
4	**D**	Each stage's critical flow ratio y = q/s; Y adds them up.
5	**D**	y = q/s says what share of the time the approach needs green to keep up.
6	**C**	C_o is the cycle length Webster's formula recommends.
7	**C**	An actuated signal extends the green while cars keep arriving and ends it when the gap between them gets too long.
8	**B**	At an isolated junction there is no green wave to keep, so SCOOT's coordination advantage mostly disappears.
9	**B**	SCOOT edits the running (elastic) plan; plan selection swaps whole stored plans.
10	**B**	Lab F3 compared mean delay per vehicle: 18.96, 14.48 and 12.79 s.

Group 6 · quiz 3 · Numbers and calculations		
1	**A**	Y = 0.60; C_o = (1.5 × 10 + 5) ÷ (1 − 0.60) = 20 ÷ 0.40 = 50 s. 33 s forgets the "1 −"; 100 s uses 2 × L.
2	**C**	Green to share = 50 − 10 = 40 s; 40 × 0.35 ÷ 0.60 = 23.3 s. 29.2 s forgets to take off L; 16.7 s is the side road.
3	**A**	Y = 0.70; C_o = (1.5 × 12 + 5) ÷ (1 − 0.70) = 23 ÷ 0.30 = 76.7 s. 23 s forgets to divide; 32.9 s divides by 0.70 instead of 0.30.
4	**D**	C_o = 20 ÷ (1 − 0.90) = 20 ÷ 0.10 = 200 s. The formula blows up as Y approaches 1.
5	**B**	(18.96 − 14.48) ÷ 18.96 = 4.48 ÷ 18.96 ≈ 0.24. 31% divides by 14.48 instead; 4.5 is the difference in seconds.
6	**B**	38.0 − 22.9 = 15.1 s; 15.1 ÷ 38.0 ≈ 0.40. 66% divides by 22.9; 1.3 s is the mean-delay gap.
7	**D**	C_o = (1.5 × 8 + 5) ÷ (1 − 0.50) = 17 ÷ 0.50 = 34 s. 17 s forgets to divide.
8	**D**	y = q ÷ s = 630 ÷ 1800 = 0.35. 2.86 inverts the ratio; 0.65 is 1 − y.
9	**A**	Green to share = 40 − 10 = 30 s; 30 × 0.20 ÷ 0.50 = 12 s. 16 s forgets to take off L; 18 s is road A.
10	**D**	Capacity = s × g ÷ C = 1800 × 18 ÷ 40 = 810 veh/h. 4000 inverts g and C.

Group 6 · quiz 4 · What happens if…?		
1	**D**	The bottom (1 − Y) shrinks towards zero, so C_o blows up: Y = 0.9, L = 10 gives 200 s.
2	**A**	Y ≥ 1 means demand needs more green than exists; real controllers cap the cycle.

3	**C**	L is on the top of (1.5 L + 5) ÷ (1 − Y), so more lost time means a longer optimum cycle.
4	**B**	A fixed plan stays good only while the flows stay the same; SCOOT adapts instead.
5	**D**	Actuation ends a green on a gap: nobody else is arriving, so the time goes to the other stage.
6	**D**	Actuation reacts only to what reaches its stop-line loops; SCOOT's upstream loops give 10–20 s of warning.
7	**D**	In light traffic SCOOT shortened the cycle; while DoS rose in the peak it held it.
8	**C**	Green is shared in proportion to flow ratio, so a larger y(A) ÷ Y means a larger share for A.
9	**D**	Rule of thumb: the more traffic changes and the closer the junctions, the more an adaptive system can help.
10	**B**	SCOOT needs working detectors and a validated model; bad inputs mean bad predicted queues and timings.

Group 6 · quiz 5 · Everyday pictures		
1	**D**	A timetable runs the same plan whatever happens: that is a fixed-time signal.
2	**C**	Actuation ends the green on a gap: no one else is arriving, so off it goes.
3	**A**	Upstream detectors give SCOOT 10–20 s of warning, so it can plan for the platoon.
4	**D**	The course says the gap is small at one junction; coordination is where SCOOT's model pays off.
5	**D**	Robot = fixed time, rabbit = actuated, dispatcher = SCOOT; the times are mean delay per vehicle.
6	**D**	C_o = (1.5 L + 5) ÷ (1 − Y).
7	**D**	Fixed time 18.96 s is the tallest bar; SCOOT 12.79 s is the shortest.
8	**A**	SCOOT beats fixed time by about a third but actuation by only about 12% at one junction.
9	**A**	SCOOT nudges the timings a few seconds at a time, following the traffic like a thermostat follows the temperature.
10	**A**	Arrivals (the tap) exceed what the fixed green drains, so the queue grows; SCOOT would give more green.

Group 6 · quiz 6 · Exam challenge		
1	**D**	C_o = 20 ÷ (1 − 0.75) = 20 ÷ 0.25 = 80 s. The stored 50 s plan is now out of date.
2	**D**	0.75 × 50 ÷ 40 = 0.9375 ≈ 0.94, above the ≈ 90% target. (At Y = 0.60 it was 0.75.)
3	**B**	Lab F3: SCOOT 12.79 s best. Corridor: actuated 29.1 s vs SCOOT 30.4 s, but SCOOT had fewer stops and less bus delay.
4	**C**	The 1.3 s mean-delay gap is small; the 15.1 s bus-delay saving is large.
5	**B**	Both actuation and SCOOT adapt; fixed time cannot, so most of the gain over fixed time comes from reacting at all.
6	**C**	Both use Robertson's platoon dispersion; TRANSYT off-line, SCOOT on-line.
7	**C**	SCOOT edits the running plan a little at a time instead of relying on one stored plan.
8	**A**	A simple controller can be enough at isolated, lightly used junctions; SCOOT needs working detectors and a validated model.
9	**C**	C_o = (1.5 × 14 + 5) ÷ 0.40 = 26 ÷ 0.40 = 65 s; green to share = 65 − 14 = 51 s.
10	**A**	Actuation reacts only locally; a green wave needs neighbours in one region sharing a cycle, with offsets set (F7).